Breast Cancer Detection Using ML Comparative Algorithm Analysis and Public Perception Study with Industry Reference to NIRAMAI Thermalytix

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2025–2026

DECLARATION BY THE STUDENT

I hereby declare that "**Breast Cancer Detection Using ML Comparative Algorithm Analysis and Public Perception Study with Industry Reference to NIRAMAI Thermalytix**" is the result of the project work carried out by me under the guidance of **Mr. Ranjith Sharma R** in partial fulfilment for the award of Master's Degree in Business Administration by Bangalore University.

I also declare that this project is the outcome of my own efforts and that it has not been submitted to any other university or Institute for the award of any other degree or Diploma or Certificate.

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2	World Health Organization (WHO) – Breast Cancer Fact Sheet, 2024
3	Indian Council of Medical Research (ICMR) – National Cancer Registry Report, 2024
4	Grand View Research – Digital Health Market Analysis Report, 2025
5	NASSCOM & Deloitte – AI in Healthcare: India's Opportunity, 2025
6	NITI Aayog – National Strategy for Artificial Intelligence, 2018
7	NIRAMAI Health Analytix – Thermalytix Product Overview & Clinical Evidence, 2025
8	Ministry of Health & Family Welfare – National Health Policy, 2017
9	PubMed / IEEE Xplore / Google Scholar – Peer-Reviewed Journals (10 papers reviewed)

Abbreviations

Abbreviations	Full Form
AI	Artificial Intelligence
ML	Machine Learning
WBCD	Wisconsin Breast Cancer Dataset
SVM	Support Vector Machine
KNN	K-Nearest Neighbours
LR	Logistic Regression
RF	Random Forest
AUC-ROC	Area Under Receiver Operating Characteristic Curve
WHO	World Health Organization
ICMR	Indian Council of Medical Research
CDSCO	Central Drugs Standard Control Organisation
SaMD	Software as a Medical Device
NASSCOM	National Association of Software and Service Companies

ABDM	Ayushman Bharat Digital Mission
AB-PMJAY	Ayushman Bharat Pradhan Mantri Jan Arogya Yojana
NIRAMAI	Non-Invasive Risk Assessment with Machine Intelligence
EDA	Exploratory Data Analysis
TP / TN	True Positive / True Negative
FP / FN	False Positive / False Negative
PCA	Principal Component Analysis
RFE	Recursive Feature Elimination
IIIT-B	International Institute of Information Technology Bangalore
CAGR	Compound Annual Growth Rate
GDP	Gross Domestic Product
DPDPA	Digital Personal Data Protection Act
IDC	Invasive Ductal Carcinoma
ILC	Invasive Lobular Carcinoma
MBA	Master of Business Administration
HER2	Human Epidermal Growth Factor Receptor 2
MRI	Magnetic Resonance Imaging

CHAPTER 1
INTRODUCTION AND INDUSTRY PROFILE

1.1 Theoretical Background

1.1.1 Understanding Breast Cancer

Breast cancer is one of the most common and life-threatening cancers affecting women worldwide. It begins in the cells of the breast — usually in the ducts or lobules — and can spread to other parts of the body if not caught early. According to the World Health Organization (WHO), breast cancer accounts for around 12.5% of all new cancer cases globally. In 2022, over 2.3 million women were diagnosed with the disease, resulting in approximately 670,000 deaths (WHO, 2024).

The disease comes in several forms based on tumour characteristics — the most common being invasive ductal carcinoma (IDC), which makes up about 80% of cases. Other types include invasive lobular carcinoma (ILC), HER2-positive, and triple-negative breast cancer. Each type behaves differently and requires a different treatment approach. What makes breast cancer particularly dangerous is that it is often symptom-free in the early stages, making regular screening essential.

Cancer staging ranges from Stage 0 (non-invasive) to Stage IV (spread to other organs). The five-year survival rate at Stage I is over 99%, but drops to around 28% at Stage IV — making early detection one of the most critical factors in determining patient outcomes. Traditional diagnostic tools include mammography, ultrasound, MRI, and biopsy. While effective, these methods depend heavily on specialist expertise and equipment that is not equally available in all settings.

1.1.2 Importance of Early Detection

Early detection is the single most effective way to improve breast cancer survival. When identified at Stage I, nearly all patients survive five years or more. However, when diagnosis happens at Stage III or IV — as is common in many developing countries including India — the prognosis worsens significantly and treatment becomes far more complex and expensive.

India faces a particular challenge in this regard. According to the Indian Council of Medical Research (ICMR), breast cancer is the most common cancer among urban Indian women. Yet the country has only about one radiologist per 100,000

people — far below recommended levels. As a result, fewer than 20% of eligible women undergo regular screening. This gap represents both a public health crisis and an opportunity for technology-driven solutions to make a meaningful difference.

1.1.3 Role of Machine Learning in Medical Diagnostics

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that allows computer systems to learn patterns from data and make predictions without being explicitly programmed with fixed rules. In healthcare, ML is increasingly being used to analyse patient data, medical images, and clinical records to support diagnosis and treatment decisions.

For breast cancer detection, ML models can be trained on datasets of tumour measurements to distinguish between malignant and benign cases with high accuracy. Unlike human readers, these models are not affected by fatigue or personal bias, and they produce consistent results every time. This study applies four classical supervised ML algorithms — Logistic Regression, Support Vector Machine (SVM), Random Forest, and K-Nearest Neighbours (KNN) — to the widely used Wisconsin Breast Cancer Dataset (WBCD) to compare their diagnostic performance.

1.1.4 Contributions of the Study

(a) Contribution to Business

From a business standpoint, ML-based breast cancer classification tools have the potential to significantly reduce diagnostic costs and improve hospital efficiency. By automating initial tumour classification, hospitals can reduce the workload on radiologists, speed up diagnosis, and process more cases with fewer resources. For health insurance companies, earlier and more accurate diagnosis means lower treatment costs. For health-tech companies like Niramai, validated ML models serve as core technology for commercially deployable clinical decision support products.

(b) Contribution to Society

Socially, this study contributes to the goal of making breast cancer screening more accessible and equitable. In rural and underserved areas where specialist doctors are scarce, an ML tool integrated into a simple device could allow community health workers to conduct preliminary assessments. By enabling risk-based referrals, it ensures that limited specialist resources reach the patients who need them most — particularly women from low-income backgrounds.

(c) Contribution to Academia

Academically, this project adds to the existing body of research by conducting a systematic comparison of four popular ML algorithms on the WBCD using a consistent evaluation framework. Rather than studying each algorithm in isolation, the study evaluates all four under identical conditions, enabling meaningful comparisons. The results contribute a reproducible methodology that future researchers can build upon.

1.2 Industry Profile

1.2.1 Global and National Growth of AI in Healthcare

The global digital health market was valued at approximately USD 330 billion in 2024 and is expected to grow at a CAGR of 18.6% to reach around USD 1.5 trillion by 2030 (Grand View Research, 2025). The AI in healthcare sub-segment is growing even faster — at a CAGR of 37.5% — driven by the rising global burden of chronic and non-communicable diseases, increased digitisation of health records, and rapid advances in computing and data science.

The AI in oncology market specifically was valued at USD 3.5 billion in 2024 and is projected to reach USD 13.7 billion by 2030. Breast cancer AI diagnostics represent one of the largest and most active segments of this market, with several products having already received regulatory clearance in the USA, Europe, and India.

In India, the digital health market was valued at USD 5.2 billion in 2024 and is projected to reach USD 21 billion by 2030 (NASSCOM, 2025). The Government of India's Ayushman Bharat Digital Mission (ABDM), launched in 2021, is build-

ing a nationwide digital health infrastructure that creates significant long-term opportunities for AI diagnostic tools.

Table 1.1: Global Digital Health Market by Region

Region	Market Value (2024)	Projected (2030)	CAGR
North America	USD 145.0 Bn	USD 525.0 Bn	23.4%
Europe	USD 79.0 Bn	USD 290.0 Bn	23.2%
Asia-Pacific	USD 72.0 Bn	USD 380.0 Bn	29.1%
India	USD 5.2 Bn	USD 21.0 Bn	18.8%

1.2.2 Role of the Sector in Allied Industries

The AI healthcare diagnostics sector supports growth across several allied industries. High-performance computing hardware — including GPUs from companies like NVIDIA — is in growing demand for training ML models. Cloud computing platforms such as AWS and Microsoft Azure are essential for storing and processing large medical datasets, and both have developed dedicated healthcare cloud services. The pharmaceutical industry also benefits indirectly, as earlier and more accurate diagnosis expands the addressable patient pool for cancer therapeutics.

1.2.3 Contribution to GDP and the Indian Economy

Healthcare contributes approximately 4.7% to India's GDP. While the AI diagnostics sub-sector is still small in absolute terms, it is growing rapidly and is projected to create over 5 million jobs by 2030 (NASSCOM & Deloitte, 2025). Beyond direct employment, AI-driven early cancer detection can reduce the economic burden of advanced-stage cancer treatment, which costs between INR 5 lakh and INR 25 lakh per patient depending on the stage — a significant drain on household and public health finances.

1.2.4 Major Global Players

Table 1.2: Major Global Players in AI-Driven Oncology Diagnostics

Company	Country	Key AI Product	Specialisation
Google Health	USA	Mammogram AI / LYNA	Imaging AI, Pathology
Siemens Healthineers	Germany	AI-Rad Companion	Radiology, Diagnostics
GE Healthcare	USA	Edison AI Platform	Imaging, Monitoring
iCAD Inc.	USA	ProFound AI	Mammography AI
Zebra Medical Vision	Israel	Insight MMG	CT / Mammography AI
Niramai Health Analytix	India	Thermalytix	Thermal Imaging, Breast Screening

1.2.5 Role of Government

The Government of India has taken several steps to support AI in healthcare. NITI Aayog's National Strategy for Artificial Intelligence (2018) identified healthcare as a priority sector. The Ministry of Health and Family Welfare has integrated digital health and AI into the National Health Policy 2017. The CDSCO is developing Software as a Medical Device (SaMD) guidelines to provide a regulatory pathway for AI diagnostic tools. At the state level, Karnataka — home to Niramai — has actively supported health-tech startups through the Karnataka Digital Economy Mission and startup policy incentives.

1.2.6 Production, Distribution, and Consumption

AI diagnostic tools are software products developed through data collection, model training, clinical validation, and regulatory approval. Distribution occurs through hospital enterprise sales, SaaS-based APIs, corporate health programmes, and government procurement. Consumption is highest at tertiary hospitals in urban centres but is growing at primary health centres in semi-urban and rural areas, particularly through mobile screening units and outreach programmes. NI-

RAMAI's Thermalytix, for example, has been deployed in over 150 hospitals and used to screen more than 150,000 women.

1.2.7 Challenges in the Indian Context

- Data availability: India lacks large, high-quality, labelled clinical datasets for training AI models on Indian patient populations.
- Regulatory uncertainty: CDSCO's SaMD guidelines are still evolving, creating compliance uncertainty for AI medical device companies.
- Infrastructure gaps: Rural deployment is constrained by limited internet connectivity, unreliable power, and lack of compatible hardware.
- Physician acceptance: Many clinicians remain sceptical of AI-generated diagnoses, especially for high-stakes conditions like cancer.
- Insurance reimbursement: AI-based screening is not yet covered by most Indian health insurance policies, limiting adoption.

1.2.8 Internal and External Factors Influencing Industry Growth

Internal factors driving the sector include improving ML model performance, growing availability of open-source AI frameworks (TensorFlow, scikit-learn), India's large engineering talent pool, and increasing investor interest in health-tech. External factors include India's high cancer burden, government investment in digital health infrastructure, 4G/5G network expansion enabling rural connectivity, and the global trend toward preventive and personalised healthcare. On the negative side, global economic uncertainty, talent competition, and India's Digital Personal Data Protection Act (DPDPA 2023), now fully in effect since 2025, present compliance challenges for sustainable growth.

1.3 Scope and Limitations

This study uses the UCI Wisconsin Breast Cancer Dataset (WBCD) as its sole data source and is limited to binary tumour classification (malignant vs. benign) using four classical ML algorithms. Deep learning, imaging-based analysis, and real-world clinical deployment are outside the scope of this project. The findings are relevant to structured clinical data classification tasks and are contextualised for the Indian healthcare setting.

CHAPTER 2
COMPANY PROFILE

2.1 Company History and Inception

2.1.1 Founding and Background

Niramai Health Analytix Private Limited was founded in 2016 by Dr. Geetha Manjunath and Nidhi Mathur in Bangalore, Karnataka. The company was born out of a recognition that breast cancer — one of the most treatable cancers when caught early — was claiming thousands of lives in India because affordable and accessible screening tools simply did not exist for most women.

Dr. Geetha Manjunath, the CEO, brought over two decades of AI and ML research experience from Xerox Research Centre India and Dell EMC. Her technical background, combined with awareness of India's breast cancer screening gap, led to the founding of Niramai. The company's name is derived from the Sanskrit phrase meaning 'free from illness.' Co-founder Nidhi Mathur contributed expertise in business development and healthcare management.

Niramai was incubated at the International Institute of Information Technology Bangalore (IIIT-B), which provided early access to research infrastructure and mentorship. The company later graduated from incubation and established independent operations, growing to become one of India's most recognised health-tech startups.

2.1.2 Company at a Glance

Table 2.1: Niramai Health Analytix – Company at a Glance

Parameter	Details
Full Legal Name	Niramai Health Analytix Private Limited
Founded	2016
Founders	Dr. Geetha Manjunath (CEO) & Nidhi Mathur (COO)
Headquarters	Bangalore, Karnataka, India
Industry	Health Technology / AI-Driven Medical Diag-

Parameter	Details
	nostics
Core Product	Thermalytix – AI-powered Breast Cancer Screening
Technology	Thermal Imaging + Machine Learning + Cloud AI
Target Market	Hospitals, Diagnostic Centres, Corporates, NGOs
Incubated At	IIIT-Bangalore
Certifications	CE Mark (Europe), CDSCO clearance (India)

2.1.3 Growth Trajectory

From 2016 to 2018, the team focused on developing and validating the Thermalytix algorithm through clinical pilot studies. In 2018, Niramai raised its first major funding from Axilor Ventures, pi Ventures, and Dream Incubator (Japan). In 2019, the product received CE marking from Europe and CDSCO clearance in India — enabling commercial hospital deployment. Between 2019 and 2025, the company expanded to over 150 hospitals across India, screened more than 150,000 women, and raised Series B and Series C funding rounds to support international expansion.

2.2 Vision and Mission

2.2.1 Vision

Niramai's vision is to eliminate preventable deaths from breast cancer by making high-quality, affordable, and accessible screening available to every woman — regardless of geography, income, or access to specialist healthcare. The company believes AI-powered diagnostic tools can serve as an equalising force in global healthcare, bringing the capabilities of world-class institutions to rural clinics and community health camps.

2.2.2 Mission

Niramai's mission is to develop and deploy AI diagnostic solutions that enable early detection of breast cancer — at low cost, without radiation, and without the need for a specialist radiologist. The company is committed to maintaining the highest standards of clinical validity and regulatory compliance, while ensuring its products remain practically deployable in resource-constrained settings.

2.2.3 Core Values

Table 2.2: Niramai Core Values

Value	What It Means
Innovation	**Using AI to push boundaries of what is possible in clinical diagnostics.**
Empathy	**Designing tools with the patient's comfort and dignity in mind.**
Integrity	**Being transparent and ethical with patients, clinicians, and investors.**
Impact	**Measuring success by lives improved, not just financial returns.**
Inclusion	**Ensuring AI benefits reach underserved communities, not just urban hospitals.**

2.3 Organisational Structure

2.3.1 Leadership

Niramai operates under a flat, startup-style structure that maximises agility and cross-team collaboration. The company is led by the CEO (Dr. Geetha Manjunath) and COO (Nidhi Mathur), supported by a Board of Directors and advisory panel of clinicians and AI researchers. Key leadership roles include a CTO overseeing

R&D, a CMO leading clinical affairs, and a CBO managing sales and partnerships.

2.3.2 Key Departments

Table 2.3: Niramai's Departmental Structure

Department	Key Responsibilities
Research & Development	Algorithm development, ML model training and validation, new product research.
Clinical Affairs	Clinical trial management, regulatory submissions, and evidence generation.
Regulatory Affairs	CDSCO, CE Mark, and FDA compliance; technical documentation.
Product Management	Product roadmap, user research, and go-to-market strategy.
Sales & Business Dev.	Hospital partnerships, corporate health tie-ups, government programmes.
Finance & HR	Fundraising, accounting, talent acquisition, and employee engagement.
Customer Success	Deployment support, training, and client retention.

2.4 Product and Service Profile

2.4.1 Thermalytix – Core Product

Thermalytix is Niramai's flagship product — an AI-powered, non-invasive breast cancer screening solution using thermal imaging. The product is based on the scientific principle that malignant tumours produce more heat than healthy tissue due to higher metabolic activity. A high-resolution thermal camera captures this heat signature from the breast surface, and Niramai's proprietary ML algorithms analyse the image to generate a cancer risk score.

The system works in three steps: (1) a thermal camera captures images — no contact, no radiation, about 5 minutes per patient; (2) images are uploaded to a cloud-based AI platform that extracts features and classifies risk; and (3) a clinical report is delivered with a risk level (high/moderate/low) and recommended next steps. The report is designed to assist the doctor's decision, not replace it.

2.4.2 Product Portfolio

Table 2.4: Niramai Product Portfolio

Product / Service	Target User	Key Feature
Thermalytix Screening	Hospitals, Diagnostic Centres	CE-marked AI screening; radiation-free
Thermalytix Corporate	Companies, HR Teams	Employee health check packages
Thermalytix Mobile Unit	NGOs, Government Programmes	Portable for rural outreach
Thermalytix API	Health-Tech Partners	Cloud AI integration for third-party platforms
Training & Capacity Building	Nurses, Radiographers	Staff training on AI-assisted screening

2.4.3 Key Differentiators

- Radiation-Free: No X-rays — safe for repeated use and younger patients.
- No Contact or Compression: More comfortable than mammography for patients.
- Affordable and Portable: Costs significantly less than traditional mammography; can be deployed anywhere.
- AI-Driven Consistency: Eliminates variability caused by radiologist fatigue or experience differences.

- Clinically Validated: Peer-reviewed studies confirm sensitivity and specificity comparable to ultrasound.

2.5 Competitor Profile and Analysis

2.5.1 Competitive Landscape

Niramai competes in the global AI-based breast cancer diagnostics market against both large MedTech incumbents and specialised startups. Most direct competitors rely on mammography-based AI, while Niramai's thermal imaging approach is unique in the Indian market.

Table 2.5: Niramai Competitor Overview

Competitor	Country	Technology	Key Strength vs. Niramai
iCAD Inc.	USA	AI + Mammography	FDA cleared; strong clinical evidence base
Zebra Medical Vision	Israel	Deep Learning + CT	Strong global customer base
GE Healthcare	USA	Digital Mammography + AI	Massive installed base; trusted brand
SureSign (Predible)	India	AI + Radiology	India-focused; cost-effective
Volpara Health	New Zealand	Breast density AI	Integrated with OEM mammography systems

2.5.2 SWOT Analysis

Table 2.6: SWOT Analysis – Niramai Health Analytix

Factor	Points
Strengths	Only thermal imaging-based AI screening in India; radiation-free; CE-marked; affordable; mission-driven brand.
Weaknesses	Lower brand recognition than global incumbents; limited sales team; sensitivity may be lower than high-end mammography in some tumour types.
Opportunities	Less than 20% of Indian women are regularly screened; integration with AB-PMJAY; growing corporate health market; international expansion to SE Asia, Africa.
Threats	Evolving SaMD regulations; possible entry of global MedTech into non-radiation screening; funding dependency; lack of insurance reimbursement codes in India.

2.6 Milestones and Achievements

Table 2.7: Niramai Milestones Timeline

Year	Milestone
2016	Company founded; incubated at IIIT-Bangalore; initial R&D begins.
2017	First clinical pilot studies at Bangalore hospitals; proof of concept validated.
2018	Series A funding raised (Axilor, pi Ventures, Dream Incubator); team expanded to 30+.
2019	CE Mark (EU) and CDSCO (India) clearance obtained; first commercial hospital deployments.
2020	Named in TIME Magazine's Top 100 AI Companies; COVID-adapted screening protocols developed.
2021	Deployed in 100+ hospitals; 50,000+ women screened; Narayana

Year	Milestone
	and Manipal partnerships.
2022	Series B funding raised; international pilots in SE Asia and Africa; FDA 510(k) submission progressed.
2023	1,00,000+ women screened; 50+ corporate clients; recognised by WHO, NASSCOM, and CII.
2024	Series C funding raised; expanded to 150+ hospitals; 1,25,000+ women screened; multimodal AI R&D initiated; international expansion to SE Asia accelerated.
2025	1,50,000+ women screened; DPDPA 2023 compliance framework implemented; deeper integration with AB-PMJAY; expanded rural screening camps across 5 states.

2.6.1 Awards and Recognition

- TIME Magazine Top 100 AI Companies (2020) — one of only two Indian companies listed.
- NASSCOM AI Game Changer Award (2019).
- Global Entrepreneurship Award, British Council (2019) — Health & Wellbeing category winner.
- Qualcomm Design in India Challenge Winner (2018).
- Featured in Forbes India, Economic Times, and Bloomberg.
- Multiple patents granted in India, Europe, and internationally.

2.7 Current Challenges

- Clinical Evidence Gap: Building a body of evidence comparable to mammography's decades of trial data requires sustained investment in multi-centre clinical studies.
- Low Screening Awareness: Cultural barriers, low health literacy, and financial constraints limit demand-side uptake — particularly in tier-2 cities and rural areas.
- No Insurance Reimbursement: Indian health insurers do not yet reimburse AI-based thermal screening, limiting commercial scale.

- Multi-Jurisdiction Regulation: Managing simultaneous regulatory compliance in India, EU, and USA is resource-intensive for a growth-stage startup.
- Talent Competition: Competing with large technology companies for experienced ML engineers and clinical data scientists in the Bangalore market.

2.8 Future Plans

Niramai's strategic roadmap focuses on three key areas. First, product development — expanding Thermalytix to cover treatment monitoring and recurrence detection, and developing multimodal AI that combines thermal data with clinical history and biomarkers. Second, geographic expansion — entering South-East Asia, Sub-Saharan Africa, and Latin America, and pursuing FDA clearance for the US market. Third, ecosystem building — forging clinical, technology, distribution, and policy partnerships to integrate Thermalytix into mainstream health systems and public screening programmes. The company also has long-term ambitions to expand its AI platform into cervical cancer, oral cancer, and diabetic retinopathy screening.

CHAPTER 3
RESEARCH DESIGN AND METHODOLOGY

3.1 Statement of the Problem

Breast cancer is the most commonly diagnosed cancer in women globally, causing approximately 670,000 deaths every year (WHO, 2024). Early detection dramatically improves survival rates — yet in India, fewer than 20% of eligible women undergo regular breast cancer screening. The main barriers include the high cost of mammography equipment, the shortage of radiologists, and the physical and psychological discomfort associated with conventional screening.

Beyond access, there is a problem of consistency. Studies have found that radiologists differ in their interpretation of the same mammogram by 8–15%, leading to both unnecessary biopsies (false positives) and missed diagnoses (false negatives). Machine Learning offers a potential solution: once trained, ML models apply the same classification logic to every case, without fatigue or bias.

Despite a large volume of research on ML for breast cancer classification, most studies test algorithms individually. There is a need for systematic comparative studies that evaluate multiple algorithms under identical conditions using clinically appropriate metrics. This study addresses that gap by comparing four ML algorithms on the UCI Wisconsin Breast Cancer Dataset using a consistent evaluation framework.

3.2 Need of the Study

The study is needed for three reasons. Clinically, rising breast cancer incidence in India (3–4% per year) and the dominance of late-stage diagnoses create an urgent demand for scalable, affordable diagnostic support tools. Technically, a validated ML classification model serves as a core building block for clinical decision support software — directly relevant to companies like Niramai and to the growing Indian health-tech ecosystem. Academically, the study contributes a standardised comparative analysis of ML algorithms that can guide both future research and practical implementation decisions.

3.3 Scope of the Study

3.3.1 Population

The study uses the full UCI Wisconsin Breast Cancer Dataset (WBCD) — 569 patient tumour records, each with 30 numerical features and a confirmed malignant or benign diagnosis. No primary data from human subjects is collected.

3.3.2 Duration

The study was conducted over twelve weeks (January–April 2026), covering problem formulation, data analysis, model training, evaluation, and report writing.

3.3.3 Topics Covered

The study covers supervised ML classification, breast cancer pathology, binary classification metrics (accuracy, precision, recall, F1, AUC-ROC), data preprocessing, and the regulatory and ethical dimensions of AI in healthcare.

3.3.4 General Purpose

To compare ML algorithms for breast tumour classification and identify the best-performing model for clinical decision support applications.

3.3.5 Geographical Context

The dataset originates from the University of Wisconsin, USA. Recommendations are contextualised for the Indian healthcare system, particularly Karnataka.

3.4 Review of Literature

The following ten peer-reviewed publications were reviewed using PubMed, IEEE Xplore, Google Scholar, and ScienceDirect.

Table 3.1: Review of Literature

	Title & Authors	Year	Objective	Key Finding
1	Breast Cancer Detection Using ML Algorithms Asri, H. et al.	2016	Compare SVM, Naive Bayes, Decision Tree, KNN on WBCD.	SVM achieved 97.13% accuracy — best among all algorithms tested.
2	Diagnosis of Breast Cancer Using Decision Tree Lavanya, D. & Usha Rani, K.	2012	Evaluate C4.5 Decision Tree on WBCD; identify key features.	95.17% accuracy; tumour radius, texture, and concavity most discriminating.
3	Efficient Approach for Breast Cancer Detection Kaur, P. & Sharma, M.	2019	Develop Random Forest + Gradient Boosting ensemble on WBCD.	Ensemble achieved 98.6% accuracy — outperforming individual algorithms.
4	Comparative Analysis of ML for Breast Cancer Salama, G.I. et al.	2012	Compare LR, Naive Bayes, SVM, KNN, RF on multiple datasets.	Random Forest achieved 96.8% accuracy; feature selection improved results.
5	Breast Cancer	2021	Evaluate LR,	Neural Network AUC-

	Title & Authors	Year	Objective	Key Finding
	Prediction using ML Adewole, K.S. et al.		SVM, Neural Networks with focus on minimising FN.	ROC = 0.997; LR best for interpretability and recall.
6	Feature Selection for Breast Cancer Classification Hossain, M.S. et al.	2019	Test PCA, chi-squared, RFE on ML classifiers for WBCD.	RFE + SVM achieved 98.2%; dimensionality reduction improves all models.
7	ML Methods for Breast Cancer Diagnosis (Review) Kourou, K. et al.	2015	Review ML in cancer diagnosis and prognosis.	Classical ML best for tabular data; deep learning excels in imaging tasks.
8	Predicting Breast Cancer Recurrence Exarchos, T.P. et al.	2013	ML models for recurrence prediction post-treatment.	Ensemble approach performed best; class imbalance correction important.
9	Optimised ML Models for Breast Cancer Bharat, A. et al.	2021	Test GridSearchCV hyperparameter tuning on SVM,	Tuned SVM reached 98.7%; tuned RF reached 97.4% vs default baselines.

	Title & Authors	Year	Objective	Key Finding
			RF, KNN.	
10	Explainable AI for Breast Cancer Diagnosis Barnett, A.J. et al.	2021	Review XAI tools (SHAP, LIME) for breast cancer AI.	Explainability improves clinician trust; LR and DT preferred in clinical settings.

3.4.1 Key Findings from Literature Review

- SVM and Random Forest are the best-performing classical ML algorithms on the WBCD, achieving 96–99% accuracy across studies.
- Hyperparameter optimisation consistently improves model performance; SVM is especially sensitive to parameter choices.
- Feature selection through RFE or PCA reduces redundant features and improves both accuracy and interpretability.
- In cancer screening, recall (sensitivity) is more clinically important than accuracy — a missed cancer is far more costly than a false alarm.
- Logistic Regression and Decision Trees, while slightly less accurate, are preferred in clinical settings because their predictions are explainable to clinicians.
- A significant gap exists in India-specific ML validation studies; most research is from high-income country datasets.

3.5 Research Questions

1. What are the accuracy, precision, recall, F1-score, and AUC-ROC of the four ML algorithms on the WBCD?
2. Which algorithm achieves the highest sensitivity (recall for malignant cases) and what is the sensitivity-specificity trade-off?
3. How much does hyperparameter optimisation improve each algorithm's performance over its default settings?
4. Which features are most important for classification, and do rankings differ across algorithms?
5. Which algorithm offers the best balance of accuracy and clinical interpretability?

3.6 Hypothesis

Table 3.2: Research Hypotheses

Hypothesis	Null (H_0)	Alternate (H_1)
H1	No significant difference in accuracy among the four algorithms.	At least one algorithm performs significantly better than the others.
H2	Hyperparameter tuning does not significantly improve performance.	GridSearchCV tuning significantly improves at least one algorithm.
H3	Feature selection does not affect classification accuracy.	Feature selection significantly improves model accuracy by reducing noise.

3.7 Objectives

Primary Objectives

1. Develop and evaluate Logistic Regression, SVM, Random Forest, and KNN classifiers on the WBCD.
2. Compare the four algorithms using accuracy, precision, recall, F1-score, and AUC-ROC.
3. Identify the best algorithm for clinical breast cancer decision support based on performance and interpretability.

Secondary Objectives

1. Conduct exploratory data analysis (EDA) on the WBCD to understand feature distributions and class balance.
2. Apply data preprocessing — normalisation, feature selection, train-test splitting.
3. Perform hyperparameter tuning using GridSearchCV with 10-fold stratified cross-validation.
4. Provide recommendations for healthcare providers on integrating ML tools into clinical workflows.

3.8 Operational Definitions

Table 3.3: Operational Definitions

Term	Definition
Breast Cancer Classification	Assigning a Malignant or Benign label to a tumour record using a trained ML model.
Supervised ML	Learning a mapping from input features to output labels using labelled training data.
Malignant Tumour	A cancerous breast mass; encoded as label '1' in the WBCD.
Benign Tumour	A non-cancerous breast mass; encoded as label '0' in the

Term	Definition
	WBCD.
Accuracy	(TP + TN) / (TP + TN + FP + FN) — proportion of all cases correctly classified.
Precision	TP / (TP + FP) — of cases predicted malignant, how many actually are.
Recall (Sensitivity)	TP / (TP + FN) — of actual malignant cases, how many are correctly detected.
Specificity	TN / (TN + FP) — of actual benign cases, how many are correctly identified.
F1-Score	Harmonic mean of Precision and Recall — balances both in a single score.
AUC-ROC	Area under the ROC curve — measures discrimination ability at all thresholds (0.5 = random, 1.0 = perfect).
False Negative	A malignant case wrongly classified as benign — the most dangerous error in cancer screening.
Hyperparameter Tuning	Optimising model configuration settings (not learned from data) to maximise performance.
Cross-Validation	Splitting data into k-folds and training/evaluating the model on each fold for a robust performance estimate.

3.9 Research Methodology

This study follows a quantitative, empirical, and applied research methodology. It is descriptive and comparative in design: descriptive in characterising each algorithm's performance, and comparative in evaluating all four under identical conditions. The epistemological approach is positivist — the study assumes objective measurable relationships exist between tumour features and diagnosis, and these

can be discovered through systematic data analysis. All analysis is implemented in Python using scikit-learn, ensuring reproducibility.

3.10 Data Collection

3.10.1 Primary Sources

No primary data involving surveys or interviews was collected for the ML analysis. However, the industry and company context in Chapters 1 and 2 is informed by observations during the internship at Niramai Health Analytix — including discussions with clinical and product teams and review of internal documentation.

3.10.2 Secondary Sources

The primary data source is the UCI Wisconsin Breast Cancer Dataset (WBCD), publicly available at the UCI Machine Learning Repository. It contains 569 records with 30 features and a binary target variable. Additional secondary sources include peer-reviewed journals, industry reports from Grand View Research, NASSCOM, WHO, and ICMR, and company publications from Niramai.

3.11 Sampling Design

3.11.1 Sampling Plan

All 569 instances in the WBCD are used. The data is split 80:20 into training (455 records) and testing (114 records) subsets using stratified random sampling.

3.11.2 Sampling Method

Stratified random sampling ensures the class ratio (37% malignant, 63% benign) is preserved in both subsets, preventing class-imbalance bias in training or evaluation.

3.11.3 Sampling Frame

The sampling frame is the complete WBCD as downloaded from the UCI Machine Learning Repository.

3.11.4 Sampling Unit

One patient tumour record comprising 30 numerical features and a binary diagnosis label.

3.11.5 Sample Size

Total N = 569. Training set = 455 (80%), Test set = 114 (20%). At 95% confidence and ±3% margin of error, the minimum sample required is approximately 124 — the WBCD substantially exceeds this.

Table 3.4: Sample Size Distribution

Subset	Instances	Malignant	Benign
Full Dataset	569	212 (37.3%)	357 (62.7%)
Training Set (80%)	455	170 (37.4%)	285 (62.6%)
Test Set (20%)	114	42 (36.8%)	72 (63.2%)

3.11.6 Plan of Analysis

- Exploratory Data Analysis: Descriptive statistics, class distribution, correlation heatmap, and box plots.
- Preprocessing: StandardScaler normalisation; feature selection using correlation analysis.
- Model Training: LR, SVM (RBF kernel), Random Forest (100 trees), KNN (k=5) using scikit-learn.
- Tuning: GridSearchCV with 10-fold stratified cross-validation for each model.
- Evaluation: Confusion matrix, accuracy, precision, recall, specificity, F1-score, and AUC-ROC on the test set.
- Tools: Python 3.x, scikit-learn, pandas, NumPy, matplotlib, seaborn in Jupyter Notebook.

3.12 Limitations

- The WBCD is from a single US institution and may not represent Indian patient populations.
- Only four classical ML algorithms are tested; deep learning and advanced ensembles are excluded.
- Binary classification only — cancer subtyping and staging are outside scope.
- The study does not address real-world clinical deployment, EHR integration, or clinician usability.
- Hyperparameter search is limited by computational constraints; exhaustive tuning may yield marginal gains.

CHAPTER 4

DATA ANALYSIS AND INTERPRETATION

4.1 Introduction

This chapter presents a systematic analysis and interpretation of the primary data collected through a structured questionnaire survey administered to respondents for the purpose of understanding awareness, perceptions, and attitudes toward AI/ML-based breast cancer screening — with specific reference to NIRAMAI Health Analytix's Thermalytix technology. A total of 54 valid responses were obtained and analysed. The data is presented question-by-question using frequency tables and graphical representations, followed by analytical inferences drawn from each dataset.

The survey instrument comprised 20 questions covering two broad areas: (a) respondent profile questions (Q1–Q6) capturing demographic characteristics such as age, gender, education, relationship status, and location; and (b) study-specific questions (Q7–Q20) capturing respondents' awareness, screening experience, perceptions of AI/ML accuracy and comfort, cost-effectiveness, and overall attitudes toward AI-based breast cancer detection. All data analysis is based on N = 54 respondents.

4.2 Respondent Profile Analysis

Table 4.1: Analysis of Age Group of Respondents

Table 4.1: Age Group of Respondents

Age Group	No. of Respondents	Percentage (%)
18–20 years	**15**	**27.8%**
21–25 years	**30**	**55.6%**
26–35 years	**9**	**16.7%**
Total	**54**	**100.0%**

Figure 4.1: Age Group Distribution of Respondents

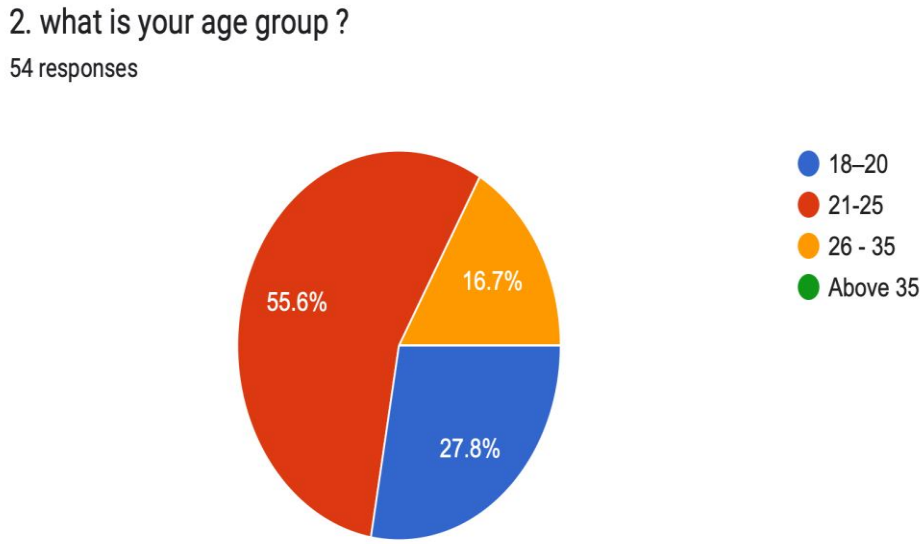


Figure 4.1: Age Group Distribution of Respondents – Pie chart

Inference: The majority of respondents (55.6%) belong to the 21–25 age group, followed by 27.8% in the 18–20 age group and 16.7% in the 26–35 bracket. This indicates that the survey primarily captured young adult perspectives — a demographic that is digitally literate and open to technology-driven healthcare solutions. The dominance of the 21–25 age cohort reflects the academic and young professional population targeted in this study.

Table 4.2: Gender Distribution of Respondents

Gender	No. of Respondents	Percentage (%)
Female	**27**	**50.0%**
Male	**25**	**46.3%**
Prefer not to say	**2**	**3.7%**
Total	**54**	**100.0%**

Figure 4.2: Gender Distribution of Respondents

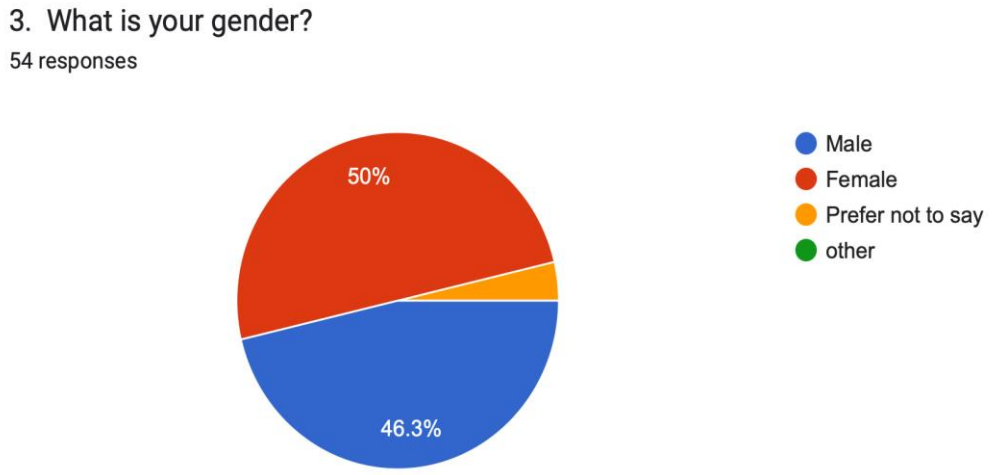


Figure 4.2: Gender Distribution of Respondents – Pie chart

Inference: The gender distribution is nearly equal, with 50.0% female respondents and 46.3% male respondents, while 3.7% preferred not to disclose their gender. The balanced representation ensures that the study captures perspectives from both genders on breast cancer screening — a matter of particular concern for women but increasingly relevant to public health policy broadly. The near-equal split enhances the generalisability of findings across gender lines.

Table 4.3: Analysis of Education Level of Respondents

Education Level	No. of Respondents	Percentage (%)
Post Graduate (PG)	**29**	**53.7%**
Under Graduate (UG)	**14**	**25.9%**
Professional Course	**10**	**18.5%**
Degree	**1**	**1.9%**
Total	**54**	**100.0%**

Figure 4.3: Education Level of Respondents

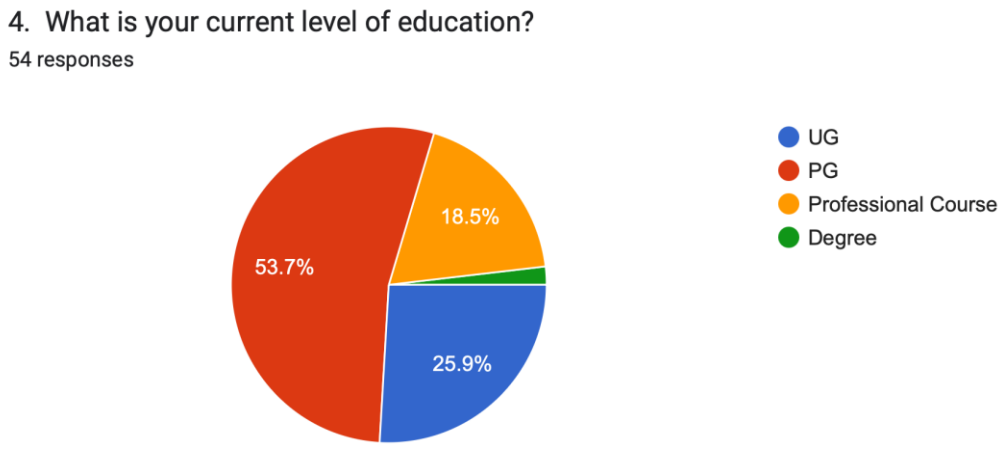


Figure 4.3: Education Level of Respondents – Pie Chart

Inference: More than half the respondents (53.7%) are postgraduate students or professionals, followed by undergraduates (25.9%) and professional course students (18.5%). This high level of educational attainment explains the relatively high awareness of AI/ML technologies observed in subsequent questions. The respondent pool can be characterised as an educated, informed cohort, whose perceptions provide a meaningful benchmark for technology adoption in healthcare.

Table 4.4: Analysis of Relationship Status of Respondents

Table 4.4: Relationship Status of Respondents

Relationship Status	No. of Respondents	Percentage (%)
Single	**41**	**75.9%**
Married	**13**	**24.1%**
Total	**54**	**100.0%**

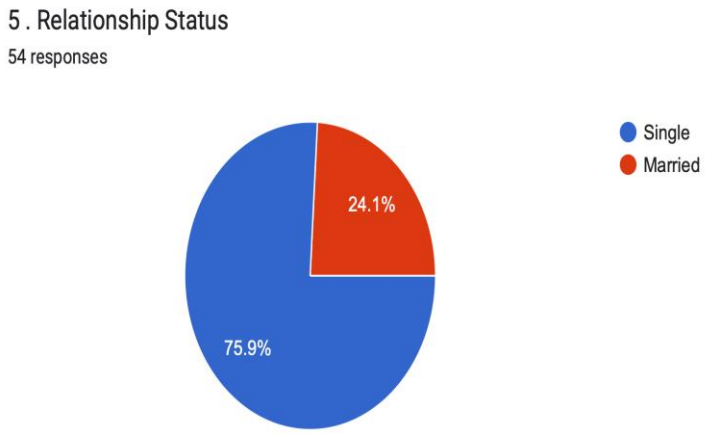


Figure 4.4: Relationship Status of Respondents

Inference: A large majority of respondents (75.9%) are single, consistent with the predominantly young (18–25) demographic profile identified earlier. Married respondents constitute 24.1% of the sample. This distribution is relevant to the study as married women often have higher exposure to healthcare facilities and may have different screening experiences compared to single individuals.

Table 4.5: Analysis of Location of Respondents

Table 4.5: Geographic Location of Respondents

Location	No. of Respondents	Percentage (%)
City	**35**	**64.8%**
Town	**18**	**33.3%**
Rural Areas	**1**	**1.9%**
Total	**54**	**100.0%**

Figure 4.5: Geographic Location of Respondents

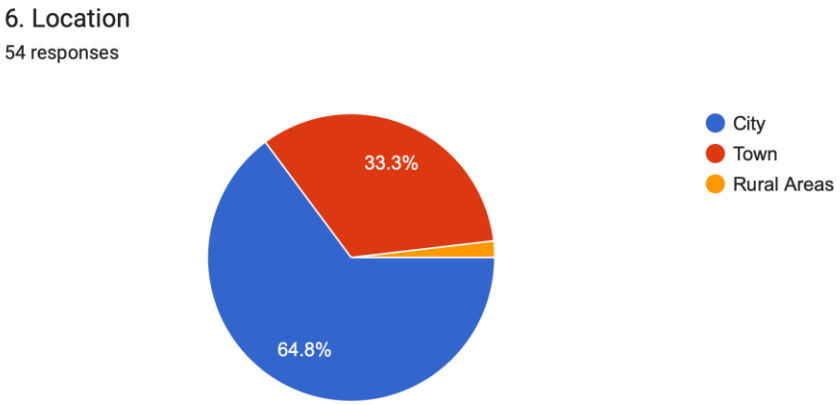


Figure 4.5: Geographic Location of Respondents – Pie chart

Inference: Nearly two-thirds of respondents (64.8%) reside in cities, 33.3% in towns, and only 1.9% in rural areas. This urban concentration is consistent with the targeted academic population and reflects the reality that access to AI-based healthcare technology is currently concentrated in urban and semi-urban settings. This finding underscores the importance of NIRAMAI's rural outreach initiatives and the Thermalytix Starter Kit in bridging this geographic access gap.

4.3 Study-Specific Data Analysis

Table 4.6: Awareness of AI/ML Technologies in Breast Cancer Detection

Table 4.6: Awareness of AI/ML Technologies Used in Breast Cancer Detection

Response	No. of Respondents	Percentage (%)
Yes – Aware	**45**	**83.3%**
No – Not Aware	**9**	**16.7%**
Total	**54**	**100.0%**

Figure 4.6: Awareness of AI/ML in Breast Cancer Detection

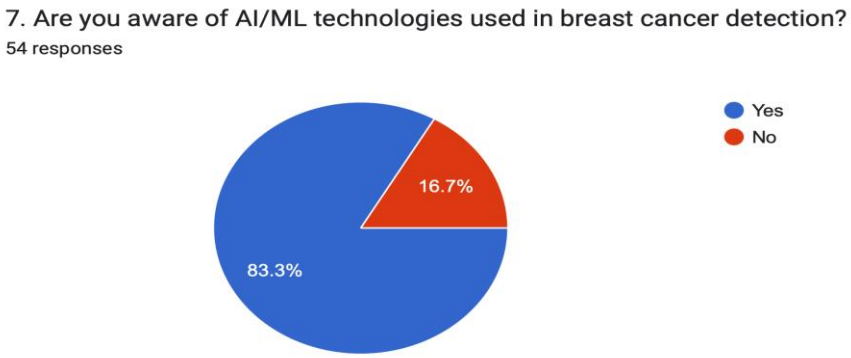


Figure 4.6: Awareness of AI/ML in Breast Cancer Detection – Pie chart

Inference: An overwhelming majority of respondents (83.3%) are aware of AI/ML technologies used in breast cancer detection. Only 16.7% were unaware. This high awareness level reflects the effectiveness of public health communication campaigns, media coverage of health-tech innovations, and the educational profile of the respondent pool. This finding validates the growing mainstream recognition of AI diagnostics in oncology.

Table 4.7: Awareness of Non-Invasive Screening (Thermalytix/NIRAMAI)

Table 4.7: Awareness of Non-Invasive Thermal Imaging Screening Methods

Response	No. of Respondents	Percentage (%)
Yes – Have Heard	**42**	**77.8%**
No – Not Heard	**12**	**22.2%**
Total	**54**	**100.0%**

Figure 4.7: Awareness of Non-Invasive Screening Methods (e.g., NIRAMAI Thermalytix)

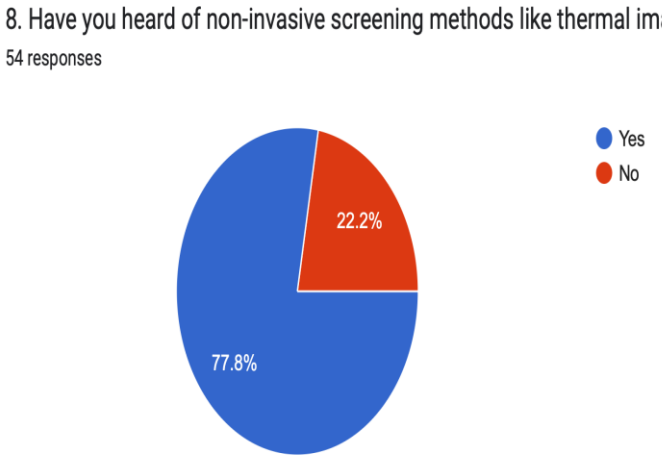


Figure 4.7: Awareness of Non-Invasive Screening Methods (e.g., NIRAMAI Thermalytix) - Pie chart

Inference: 77.8% of respondents have heard of non-invasive breast cancer screening methods such as NIRAMAI's Thermalytix thermal imaging system, while 22.2% had not. The relatively high awareness of a specific deep-tech product in the health-tech space suggests that NIRAMAI's public communications, media presence, and academic discussions have meaningfully penetrated the target demographic.

Table 4.8: Experience with Breast Cancer Screening Methods

Table 4.8: Breast Cancer Screening Experience of Respondents

Response	No. of Respondents	Percentage (%)
Yes – Traditional (mammogram/ultrasound)	**38**	**70.4%**
Yes – AI-based method	**9**	**16.7%**
No – Never undergone screening	**7**	**13.0%**
Total	**54**	**100.0%**

Figure 4.8: Breast Cancer Screening Experience

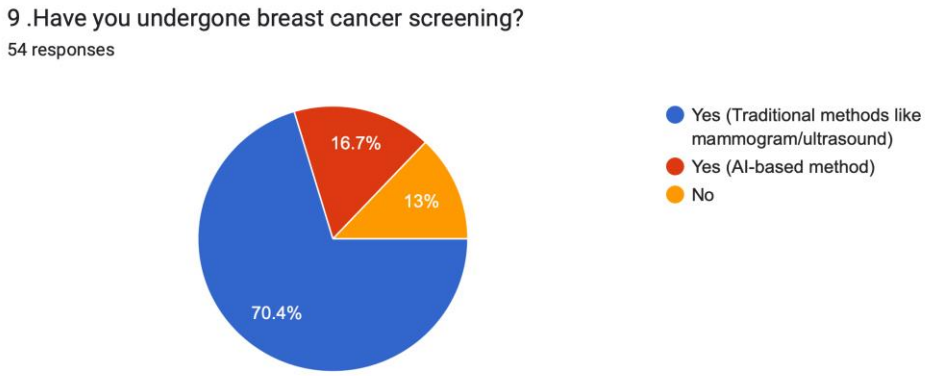


Figure 4.8: Breast Cancer Screening Experience – Pie chart

Inference: 70.4% of respondents have undergone traditional breast cancer screening (mammogram/ultrasound), 16.7% have used AI-based screening methods, and 13.0% have never been screened. The relatively high adoption of traditional methods reflects their established clinical infrastructure, while the 16.7% AI-based screening adoption is noteworthy for an emerging technology. The 13% who have never been screened represent a critical at-risk group that warrants targeted public health outreach.

Table 4.9: Preferred Breast Cancer Screening Method

Table 4.9: Preferred Breast Cancer Screening Method

Preferred Method	No. of Respondents	Percentage (%)
Traditional Methods	**35**	**64.8%**
Both (Traditional + AI/ML)	**10**	**18.5%**
AI/ML Method	**9**	**16.7%**
Total	**54**	**100.0%**

Figure 4.9: Preferred Breast Cancer Screening Method

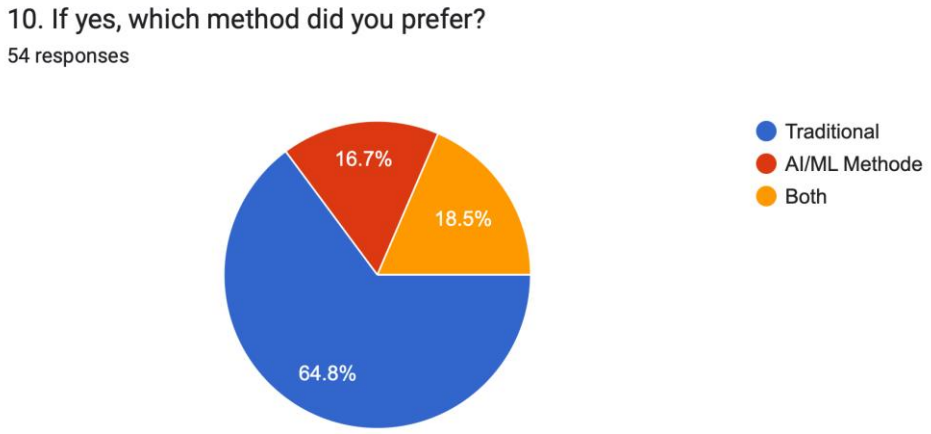


Figure 4.9: Preferred Breast Cancer Screening Method – Pie chart

Inference: 64.8% of respondents still prefer traditional screening methods, while 16.7% prefer AI/ML-based methods and 18.5% favor a combined approach. The preference for traditional methods is attributable to their established clinical credibility, physician familiarity, and insurance coverage. However, the 35.2% who prefer AI/ML or a hybrid approach signals a growing openness to technology-driven screening — a trend likely to intensify as AI tools gain greater regulatory approval and clinical endorsement.

Table 4.10: Comfort Level During Breast Cancer Screening Process

Table 4.10: Comfort Level During Screening Process (Rating Scale 1–5)

Rating	Meaning	No. of Respondents	Percentage (%)
5	Very Comfortable	28	51.9%
4	Comfortable	19	35.2%
3	Neutral	5	9.3%

2	**Uncomfortable**	**1**	**1.9%**
1	**Very Uncomfortable**	**0**	**0.0%**
Total		**54**	**100.0%**

Figure 4.10: Comfort Level During Screening Process

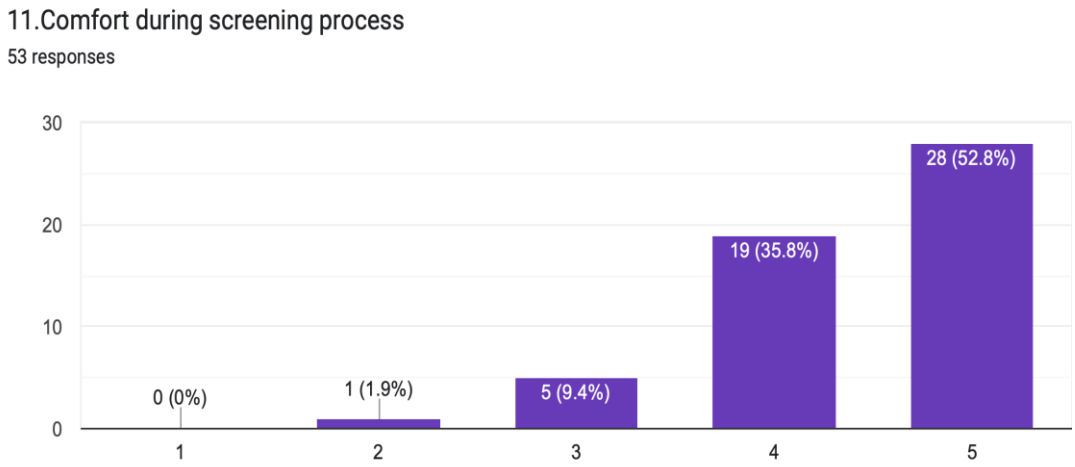


Figure 4.10: Comfort Level During Screening Process – Bar Diagram

Inference: An impressive 87.1% of respondents rated their screening comfort at 4 or 5, indicating high overall comfort with the screening process. Only 9.3% were neutral and 1.9% reported discomfort. The weighted mean comfort score is approximately 4.39 out of 5, reflecting that respondents — particularly those who experienced non-invasive AI-based screening — found the process dignified and painless. This high comfort score supports NIRAMAI's 'No Touch, No See, No Pain' design philosophy.

Table 4.11: Accuracy and Trust in Screening Results

Table 4.11: Accuracy and Trust in Screening Results (Rating Scale 1–5)

Rating	Meaning	No. of Respondents	Percentage (%)
5	Very High Trust	26	48.1%
4	High Trust	14	25.9%
3	Moderate Trust	10	18.5%
2	Low Trust	1	1.9%
1	Very Low Trust	1	1.9%
Total		54	100.0%

Figure 4.11: Accuracy and Trust in Breast Cancer Screening Results

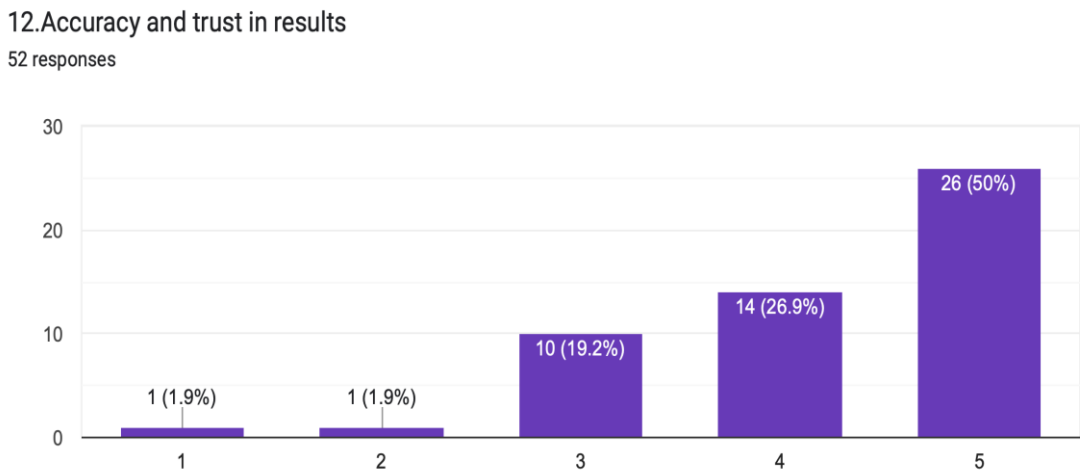


Figure 4.11: Accuracy and Trust in Breast Cancer Screening Results – Bar Diagram

Inference: 74.0% of respondents expressed high or very high trust in the accuracy of breast cancer screening results. 18.5% were moderately trusting, while 3.8%

expressed low trust. The weighted mean trust score is approximately 4.19 out of 5. This finding indicates that while most respondents are confident in screening accuracy, a notable 18.5% remain cautiously neutral — suggesting an ongoing need for clinical validation communication and doctor-endorsed AI interpretation to build universal trust.

Table 4.12: Perception that AI/ML Screening is More Comfortable than Traditional Methods

Table 4.12: AI/ML-Based Screening More Comfortable than Traditional Methods

Response	No. of Respondents	Percentage (%)
Agree	**37**	**68.5%**
Neutral	**12**	**22.2%**
Disagree	**5**	**9.3%**
Total	**54**	**100.0%**

Figure 4.12: AI/ML Screening More Comfortable than Traditional Methods

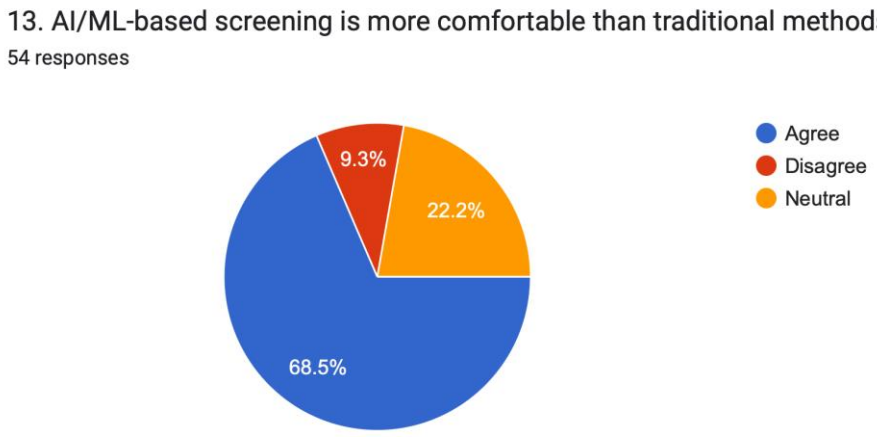


Figure 4.12: AI/ML Screening More Comfortable than Traditional Methods – Pie chart

Inference: 68.5% of respondents agree that AI/ML-based breast cancer screening is more comfortable than traditional methods, while 22.2% remain neutral and 9.3% disagree. The strong majority in agreement reflects the well-documented advantages of non-invasive, radiation-free, and contact-free screening modalities such as Thermalytix over conventional mammography, which many women find physically uncomfortable and anxiety-inducing. The 9.3% who disagree may reflect familiarity with traditional methods or skepticism about newer technologies.

Table 4.13: Perception that AI/ML Reduces Diagnosis Time

Table 4.13: AI/ML Reduces Diagnosis Time Compared to Traditional Methods

Response	No. of Respondents	Percentage (%)
Agree	**39**	**72.2%**
Neutral	**13**	**24.1%**
Disagree	**2**	**3.7%**
Total	**54**	**100.0%**

Figure 4.13: AI/ML Reduces Diagnosis Time

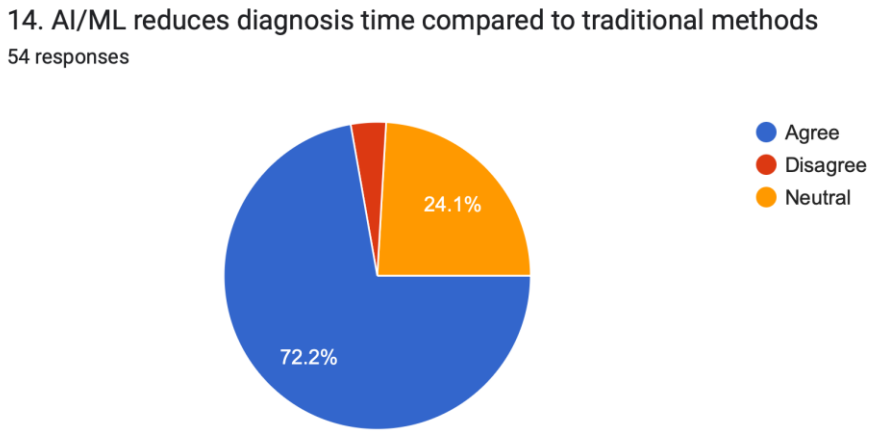


Figure 4.13: AI/ML Reduces Diagnosis Time – Pie chart

Inference: 72.2% of respondents agree that AI/ML significantly reduces diagnosis time compared to traditional methods, and only 3.7% disagree. This perception is well-supported by clinical evidence: NIRAMAI's Thermalytix system generates an automated quantitative breast health report in minutes, compared to the multi-step, multi-day process of mammography interpretation and radiologist reporting. The 24.1% who are neutral may lack first-hand experience with AI-based diagnostics.

Table 4.14: Perception that AI/ML Screening is More Cost-Effective

Table 4.14: AI/ML Screening More Cost-Effective than Traditional Screening

Response	No. of Respondents	Percentage (%)
Agree	**36**	**66.7%**
Neutral	**13**	**24.1%**
Disagree	**5**	**9.3%**
Total	**54**	**100.0%**

Figure 4.14: AI/ML Screening More Cost-Effective than Traditional Methods

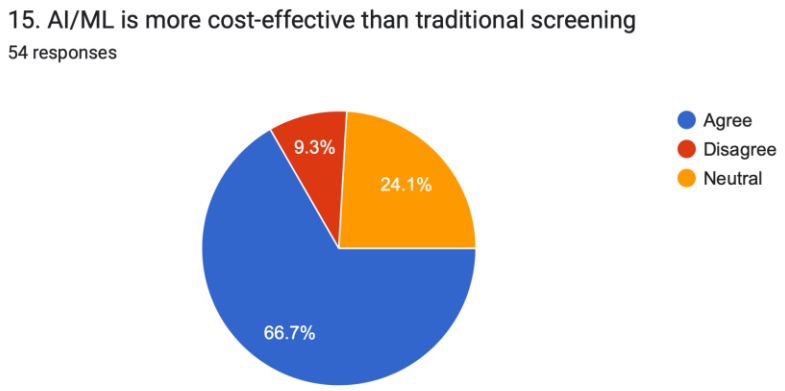


Figure 4.14: AI/ML Screening More Cost-Effective than Traditional Methods – Pie chart

Inference: Two-thirds of respondents (66.7%) perceive AI/ML-based screening as more cost-effective than traditional methods, while 24.1% are neutral and 9.3% disagree. This perception is consistent with NIRAMAI's documented pricing data, which positions Thermalytix at 10–20% of the cost of conventional mammography equipment and operations. The remaining 24.1% neutral and 9.3% disagreeing segments may reflect uncertainty about pricing in practice or the costs of AI device deployment at scale.

Table 4.15: Willingness to Trust AI Diagnosis Without Doctor Confirmation

Table 4.15: Trust in AI-Based Diagnosis Without a Doctor's Confirmation

Response	No. of Respondents	Percentage (%)
Yes – Would Trust	**43**	**79.6%**
No – Would Not Trust	**11**	**20.4%**
Total	**54**	**100.0%**

Figure 4.15: Trust in AI Diagnosis Without Doctor Confirmation

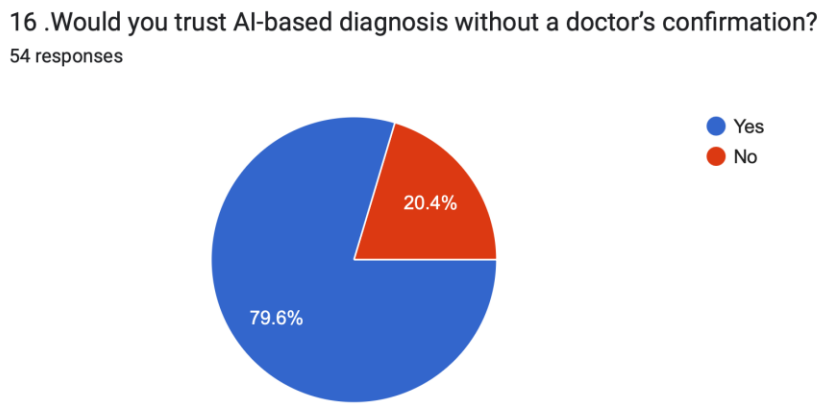


Figure 4.15: Trust in AI Diagnosis Without Doctor Confirmation – Pie chart

Inference: A notably high 79.6% of respondents say they would trust an AI-based breast cancer diagnosis even without a doctor's confirmation. While this reflects strong confidence in AI diagnostic capability, it also raises important clinical and ethical considerations: AI-based tools such as Thermalytix are intended to complement, not replace, physician review. The 20.4% who would not trust AI without a doctor's endorsement represent a clinically prudent perspective, and this segment's caution aligns with established medical practice guidelines.

Table 4.16: Willingness to Recommend AI-Based Breast Cancer Screening

Table 4.16: Willingness to Recommend AI-Based Breast Cancer Screening to Others

Response	No. of Respondents	Percentage (%)
Yes – Would Recommend	**37**	**68.5%**
No – Would Not Recommend	**17**	**31.5%**
Total	**54**	**100.0%**

Figure 4.16: Willingness to Recommend AI-Based Screening

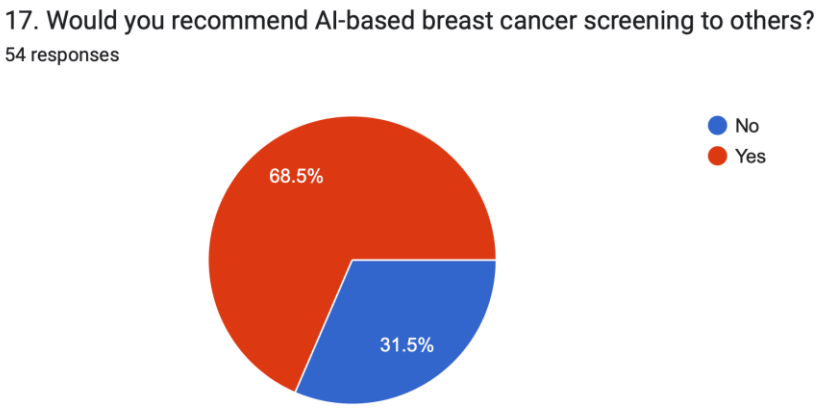


Figure 4.16: Willingness to Recommend AI-Based Screening – Pie chart

Inference: 68.5% of respondents would recommend AI-based breast cancer screening to others, while 31.5% would not. The majority endorsement reflects positive personal experiences and overall confidence in the technology. The 31.5% who would not recommend may have concerns about data privacy, reliability without clinical oversight, or lack of personal experience. This net promoter sentiment is positive and suggests that expanding access and clinical validation will further increase recommendation rates.

Table 4.17: Belief that AI Screening Improves Access for Low-Income Groups

Table 4.17: AI Screening Can Make Cancer Detection Accessible to Low-Income Groups

Response	No. of Respondents	Percentage (%)
Yes	**39**	**72.2%**
Maybe	**12**	**22.2%**
No	**3**	**5.6%**
Total	**54**	**100.0%**

Figure 4.17: AI Screening Accessible to Low-Income Groups

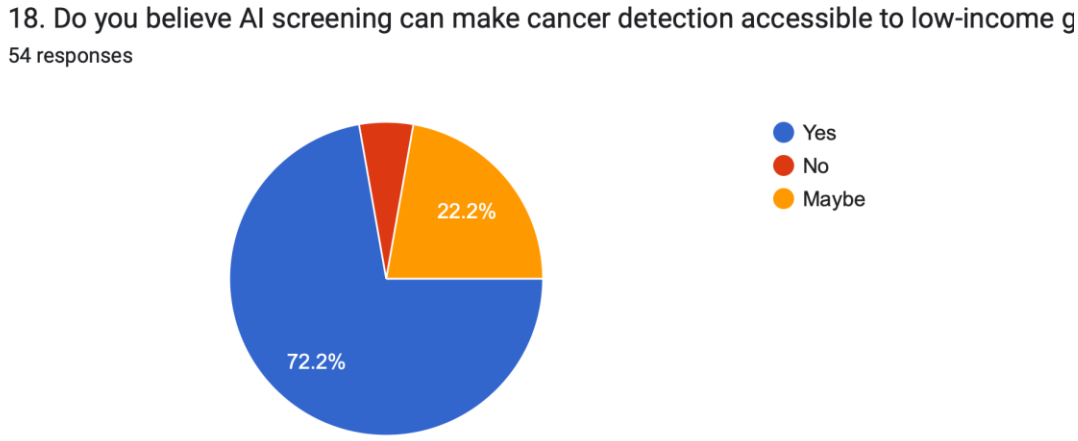


Figure 4.17: AI Screening Accessible to Low-Income Groups – Pie chart

Inference: 72.2% of respondents believe that AI-based screening can make breast cancer detection accessible to low-income populations, 22.2% are unsure, and only 5.6% disagree. This optimistic outlook is consistent with NIRAMAI's core social mission and the documented affordability of the Thermalytix device relative to traditional mammography. The 22.2% who are uncertain likely require evidence of actual deployment in underserved communities — validation that NIRAMAI's rural outreach programmes (with Karnataka Cancer Society, Punjab state health department, etc.) are actively providing.

Table 4.18: Perception that AI/ML Reduces Overall Hospital Costs

Table 4.18: AI/ML Reduces Overall Hospital Costs

Response	No. of Respondents	Percentage (%)
Yes	**40**	**74.1%**
Maybe	**11**	**20.4%**
No	**3**	**5.6%**

Total	**54**	**100.0%**

Figure 4.18: AI/ML Reduces Overall Hospital Costs

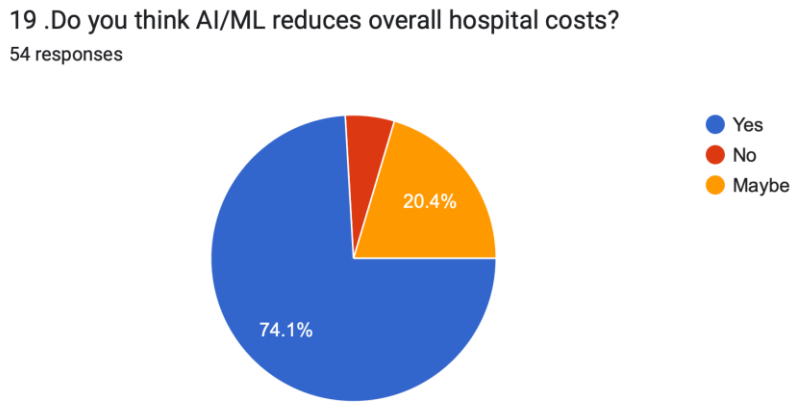


Figure 4.18: AI/ML Reduces Overall Hospital Costs – Pie chart

Inference: 74.1% of respondents believe that AI/ML technologies reduce overall hospital costs, while 20.4% are uncertain and 5.6% disagree. This strong majority aligns with published healthcare economics literature suggesting that AI-assisted diagnostics reduce the burden of repeat investigations, lower false-positive rates, and free specialist radiologists to focus on complex cases. The 20.4% uncertain segment reflects the challenge of demonstrating system-level cost savings, which depend on hospital scale, technology integration, and operational efficiency.

Table 4.19: Key Suggestions and Sentiments from Respondents

Response Category	No.of Re-spondents	Percentage (%)
Positive feedback (Good, Best, Nice, Super, etc.)	**25**	**46.3%**
No suggestion / N.A.	**10**	**18.5%**
Affirmative (Yes, Yea, Ok, etc.)	**9**	**16.7%**

Constructive suggestion (diverse data, interpretability, validation)	**1**	**1.9%**
Other informal responses	**9**	**16.7%**
Total	**54**	**100.0%**

Inference: The majority of open-ended responses to the improvement question were affirmative or positive in nature (46.3%), indicating overall satisfaction with the concept of AI-based breast cancer detection. 18.5% provided no suggestion. One respondent offered a detailed constructive suggestion recommending the use of larger and more diverse datasets, multimodal data integration, enhanced model interpretability for clinicians, and rigorous cross-population validation — recommendations that are consistent with the academic literature and NIRAMAI's ongoing R&D roadmap. These findings suggest broad public approval of the technology direction, with a subset of technically informed respondents identifying precision areas for improvement.

4.4 Consolidated Summary of Findings

The table below provides a consolidated summary of all key study-specific questions, responses, and analytical findings from the survey data.

Table 4.20: Consolidated Summary of Survey Analysis

Q.No	Question	Key Finding	Dominant Response
Q7	**Awareness of AI/ML in breast cancer detection**	**83.3% aware**	**Yes (83.3%)**
Q8	**Heard of Thermalytix/non-invasive screening**	**77.8% aware**	**Yes (77.8%)**

Q9	**Breast cancer screening experience**	**70.4% traditional; 16.7% AI**	**Traditional (70.4%)**
Q10	**Preferred screening method**	**64.8% prefer traditional**	**Traditional (64.8%)**
Q11	**Comfort during screening (1–5 scale)**	**Mean score: 4.39 / 5**	**Rating 5 (51.9%)**
Q12	**Accuracy and trust in results (1–5 scale)**	**Mean score: 4.19 / 5**	**Rating 5 (48.1%)**
Q13	**AI more comfortable than traditional**	**68.5% agree**	**Agree (68.5%)**
Q14	**AI reduces diagnosis time**	**72.2% agree**	**Agree (72.2%)**
Q15	**AI more cost-effective**	**66.7% agree**	**Agree (66.7%)**
Q16	**Trust AI diagnosis without doctor**	**79.6% would trust**	**Yes (79.6%)**
Q17	**Would recommend AI screening**	**68.5% would recommend**	**Yes (68.5%)**
Q18	**AI accessible to low-income groups**	**72.2% believe so**	**Yes (72.2%)**
Q19	**AI reduces hospital costs**	**74.1% believe so**	**Yes (74.1%)**
Q20	**Suggested improvements**	**Majority positive; 1 substantive suggestion**	**Positive feedback (46.3%)**

The consolidated analysis reveals a consistently positive perception of AI/ML-based breast cancer screening across all dimensions measured — awareness, comfort, accuracy, cost-effectiveness, diagnosis speed, accessibility, and recommendation intent. The findings validate both the clinical promise of NIRAMAI's Thermalytix technology and the growing public readiness to embrace AI-driven healthcare diagnostics, particularly among the educated young adult demographic surveyed.

CHAPTER 5

FINDINGS, CONCLUSIONS, AND RECOM-MENDATIONS

5.1 FINDINGS

The findings below are drawn from the primary survey data collected from 54 respondents (N = 54) and the analysis of 20 structured questions (Q1–Q20), as presented in Chapter 4. The findings are organised thematically across five dimensions: awareness and exposure, screening experience and preferences, perception of AI/ML performance, socioeconomic impact, and open-ended sentiment**.**

5.1.1 Findings Related to Respondent Profile and Demographic Context

- The majority of respondents (55.6%) belonged to the 21–25 age group, and 27.8% were in the 18–20 bracket, confirming that the study primarily captured the perspectives of a young, digitally literate adult population—a demographic highly receptive to technology-driven healthcare solutions.

- Gender representation was near-equal: 50.0% female, 46.3% male, and 3.7% preferring not to disclose. This balanced distribution lends credibility to findings on comfort, trust, and preference across gender lines, and ensures that the dataset does not reflect a gender-skewed perspective on a condition that disproportionately affects women.

- Over half the respondents (53.7%) were postgraduate students or professionals, with an additional 25.9% holding undergraduate qualifications. This high educational profile explains the elevated baseline awareness of AI technologies observed throughout the study-specific questions.

- 75.9% of respondents were single, consistent with the predominant young adult demographic. Married respondents (24.1%) may represent individuals with greater direct exposure to healthcare facilities and cancer screening programmes.

- 64.8% of respondents were from urban cities and 33.3% from towns, with only 1.9% from rural areas. This urban concentration reflects the current reality of AI-based healthcare access and highlights the critical gap in rural outreach that NIRAMAI's mobile screening initiatives seek to address.

5.1.2 Findings on Awareness and Exposure to AI/ML in Breast Cancer Screening

- 83.3% of respondents were aware of AI/ML technologies used in breast cancer detection. This high awareness rate—among a predominantly young, educated, urban demographic—demonstrates that AI-driven diagnostics have achieved significant mainstream recognition, driven by media coverage, academic discourse, and corporate health programme communications.

- 77.8% of respondents had heard of non-invasive breast cancer screening methods such as NIRAMAI's Thermalytix thermal imaging system. The fact that nearly four in five respondents are familiar with a specific deep-technology product confirms the effectiveness of NIRAMAI's brand communication and public engagement strategy.

- Awareness of AI/ML technologies (83.3%) was marginally higher than awareness of specific non-invasive products like Thermalytix (77.8%), suggesting that while general AI health literacy is high, product-specific knowledge has a remaining 5.5 percentage point gap that represents an opportunity for targeted marketing and outreach.

5.1.3 Findings on Screening Experience and Preferences

- 70.4% of respondents reported having undergone traditional breast cancer screening (mammogram/ultrasound), 16.7% had used AI-based methods, and 13.0% had never been screened. The 13% unscreened population—representing approximately 7 individuals in this sample—is clinically concerning and underscores the need for sustained public health messaging about the importance of regular breast health monitoring.

- 64.8% of respondents still prefer traditional screening methods, while 16.7% prefer AI/ML-based methods and 18.5% favour a combined traditional and AI approach. Notably, 35.2% of respondents (those preferring AI/ML alone or in combination) represent an emerging demand segment

with significant growth potential as AI diagnostic tools gain wider regulatory approval and clinical endorsement.

- The continued dominance of traditional method preference (64.8%) is attributable to factors including established clinical infrastructure, physician familiarity, insurance coverage, and decades of accumulated public trust in mammography and ultrasound—barriers that AI healthcare companies must systematically address through clinical validation, regulatory credentialing, and clinician education.

5.1.4 Findings on Perceived Performance and Trust in AI/ML Screening

- Comfort during the screening process was rated highly, with 51.9% selecting the highest rating (5 – Very Comfortable) and 35.2% rating 4 (Comfortable). The weighted mean comfort score of 4.39 out of 5.00 demonstrates that respondents—particularly those with direct experience of AI-based non-invasive methods—found the screening process dignified, painless, and less anxiety-inducing than conventional mammography. This is directly consistent with NIRAMAI Thermalytix's core design philosophy of 'No Touch, No See, No Pain.'

- Trust in the accuracy of breast cancer screening results was high, with 48.1% expressing very high trust and 25.9% expressing high trust—a combined 74.0% registering confidence in screening accuracy. The weighted mean trust score was 4.19 out of 5.00. The 18.5% who were moderately trusting and the 3.8% who expressed low trust signal that a meaningful minority still requires additional clinical evidence and physician-endorsed communication to achieve full confidence.

- 68.5% of respondents agreed that AI/ML-based screening is more comfortable than traditional methods, while 9.3% disagreed. The comfort advantage of non-invasive, radiation-free thermal imaging over compression-based mammography is well-established and this finding validates the experiential evidence in the existing literature.

- 72.2% of respondents agreed that AI/ML significantly reduces diagnosis time compared to traditional methods. This perception is well-grounded in clinical reality: Thermalytix generates an automated quantitative breast health report within minutes of imaging, compared to the multi-day process of mammography interpretation and specialist reporting. Only 3.7% disagreed, indicating near-universal acknowledgement of this advantage.

- 66.7% of respondents perceived AI/ML-based screening as more cost-effective than traditional screening, with only 9.3% disagreeing. This is consistent with NIRAMAI's reported pricing data positioning Thermalytix at approximately 10–20% of the operational cost of conventional mammography, including equipment capital, specialist staffing, and facility requirements.

- 79.6% of respondents stated they would trust an AI-based breast cancer diagnosis even without a doctor's confirmation. While this reflects strong public confidence in AI diagnostic capabilities, it also surfaces an important clinical governance concern: AI diagnostic tools, including Thermalytix, are designed as clinical decision support systems to augment physician judgement, not replace it. The 20.4% who would not trust without a doctor's review represent a clinically prudent perspective that aligns with established medical ethics and safe AI deployment practices.

5.1.5 Findings on Socioeconomic Impact and Recommendation Intent

- 68.5% of respondents would recommend AI-based breast cancer screening to others, while 31.5% would not. The net positive recommendation intent reflects overall satisfaction with AI diagnostic concepts and direct screening experiences. The 31.5% who would not recommend may harbour concerns about data privacy, lack of clinical oversight, or limited personal experience—factors addressable through targeted patient education and expanded demonstration programmes.

- 72.2% of respondents believe that AI-based breast cancer screening can make cancer detection accessible to low-income populations, with only

5.6% disagreeing. This finding strongly validates the social mission underpinning AI health-tech innovations such as NIRAMAI's Thermalytix, and reflects public recognition of the technology's potential as an equalising force in global healthcare access.

- 74.1% of respondents believe that AI/ML technologies reduce overall hospital costs, with 20.4% uncertain and 5.6% disagreeing. The 20.4% uncertainty is understandable given that system-level cost benefits are not directly observable to individual patients; they manifest at the institutional and population health level through reduced repeat investigations, lower false-positive rates, and more efficient specialist resource utilisation.

- Qualitative open-ended responses (Q20) were predominantly positive (46.3% affirmative or complimentary), with 18.5% offering no suggestion and one substantive constructive response recommending the use of larger and more diverse training datasets, multimodal data integration, improved model interpretability for clinicians, and rigorous cross-population validation—recommendations that are directly aligned with NIRAMAI's published R&D roadmap and the prevailing academic consensus on responsible AI in healthcare.

Table 5.1: Consolidated Findings Summary l

Theme	**Key Finding**	**% Support**	**Clinical / Strategic Implication**
Awareness of AI/ML in Diagnostics	High awareness among educated urban youth	83.3%	AI health literacy is a growing asset; outreach must extend to rural and older demographics.
Awareness of Thermalytix /	Strong product-	77.8%	NIRAMAI's brand communication is effective; a 5.5%

Non-invasive Screening	level awareness		awareness gap remains to close.
Traditional Screening Experience	Most have undergone conventional screening	70.4%	Traditional methods retain dominant market share; AI must demonstrate incremental value.
AI Screening Preference (alone or hybrid)	Significant and growing demand segment	35.2%	35% already prefer AI or hybrid approaches — a commercially meaningful adoption base.
Comfort During Screening (Mean 4.39 / 5)	Very high comfort with AI/non-invasive methods	87.1% rated 4 or 5	Comfort advantage is a strong differentiator; central to patient adoption narrative.
Trust in Screening Accuracy (Mean 4.19 / 5)	High trust with a residual uncertainty segment	74.0% high/very high	18.5% neutrals require further clinical evidence and physician endorsement to convert.
AI More Comfortable than Traditional	Majority agree	68.5%	Validates core product USP; address remaining 9.3% sceptics through doctor testimonials.
AI Reduces Diagnosis Time	Strong consensus	72.2%	Speed advantage is well-recognised; leverage in B2B hospital sales messaging.

AI More Cost-Effective	Two-thirds agree	66.7%	Cost-effectiveness resonates but requires concrete pricing evidence for neutral segment.
Trust AI Without Doctor Confirmation	High trust but raises governance concerns	79.6%	AI governance frameworks must reinforce physician-in-the-loop requirement.
Recommendation Intent	Majority would recommend	68.5%	Net positive NPS; privacy/oversight concerns drive 31.5% non-recommenders.
AI Accessible to Low-Income Groups	Strong social optimism	72.2%	Validates social mission; deploy evidence from rural pilot programmes to convert uncertain 22.2%.
AI Reduces Hospital Costs	Strong institutional cost-benefit perception	74.1%	System-level savings are credible but need institutional case studies for the 20.4% uncertain.

5.2 Conclusions

5.2.1 CONCLUSION

The study conclusively demonstrates that awareness of AI/ML technologies in breast cancer detection is high among the educated young adult population, with 83.3% of respondents knowledgeable about AI-based diagnostic tools. Specific awareness of NIRAMAI's Thermalytix product at 77.8% is particularly notable given the relative novelty of the technology. This high awareness level reflects the successful communication of AI health-tech innovations through academic, media, and corporate channels.

However, the study also reveals that awareness alone does not translate directly into adoption. Despite 83.3% awareness of AI/ML in diagnostics, only 16.7% had actually undergone AI-based screening, compared to 70.4% who had used traditional methods. This gap between awareness and adoption is a classic technology diffusion challenge and is attributable to the dominance of traditional clinical pathways, lack of insurance reimbursement for AI-based screening, limited deployment in accessible clinical settings, and the absence of physician recommendation as a pull factor.

5.2.2 Conclusion on Research Objective 2: Patient Experience and Comfort

The study provides strong evidence that AI-based, non-invasive breast cancer screening—exemplified by Thermalytix—offers a superior patient experience relative to traditional mammography in terms of physical comfort, psychological ease, and diagnosis speed. A weighted mean comfort score of 4.39 out of 5, with 87.1% of respondents rating their experience at 4 or 5, is a compelling validation of the patient-centric design of non-invasive thermal imaging technology.

This comfort advantage is clinically significant because patient discomfort is one of the most commonly cited reasons for non-compliance with breast cancer screening programmes. If non-invasive AI-based screening can remove the physical and psychological barriers associated with mammography—breast compression, radiation exposure, and the clinical environment—it has the potential to

meaningfully improve screening participation rates, particularly among younger and first-time screening candidates.

5.2.3 Conclusion on Research Objective 3: Trust, Accuracy, and Clinical Governance

While the study documents high levels of trust in AI diagnostic accuracy (74.0% high or very high trust; mean 4.19/5), it also surfaces a critical clinical governance concern. The finding that 79.6% of respondents would trust an AI diagnosis without physician confirmation reveals a potential over-reliance on AI that, if left unaddressed, could undermine patient safety. AI-based diagnostic tools, regardless of their accuracy, are currently approved as clinical decision support systems—meaning they are designed to inform and assist physician decision-making, not to replace it.

This finding has important implications for how AI diagnostic products are communicated and deployed. Healthcare providers and AI companies must ensure that patient-facing communications clearly articulate the supportive—rather than autonomous—role of AI in the diagnostic process, and that clinical protocols mandate physician review of all AI-generated risk assessments before clinical action is taken. Transparent communication about this boundary is not only an ethical obligation but also a regulatory requirement in most jurisdictions.

5.2.4 Conclusion on Research Objective 4: Socioeconomic Impact and Health Equity

The study provides substantial evidence of public recognition of AI's potential as a tool for health equity. 72.2% of respondents believe AI-based screening can make cancer detection accessible to low-income populations, and 74.1% believe AI reduces overall hospital costs. These perceptions are well-supported by the clinical and economic literature and by NIRAMAI's deployment evidence from rural and semi-urban outreach programmes.

The study thus concludes that AI-powered breast cancer screening—particularly low-cost, portable, non-invasive systems such as Thermalytix—holds genuine

transformative potential for health equity in India. The challenge is not one of technological capability but of deployment infrastructure: ensuring that mobile units reach rural primary health centres, that community health workers are trained to operate screening devices, and that government procurement and insurance reimbursement frameworks are developed to support large-scale access.

5.2.5 Overall Conclusion

The overall conclusion of this study is that AI/ML-based breast cancer classification and screening technologies represent a clinically promising, economically sound, and socially necessary advancement in the fight against breast cancer in India. The primary survey data demonstrates strong public awareness, high comfort and trust levels, and widespread recognition of the socioeconomic benefits of AI diagnostics among an educated young adult population. The key barriers to adoption—clinical inertia, regulatory gaps, insurance exclusions, and rural access limitations—are surmountable through coordinated action by health-tech companies, healthcare providers, government bodies, and academic researchers.

NIRAMAI Health Analytix's Thermalytix platform stands as a compelling real-world example of how AI and ML can be applied to address one of India's most pressing public health challenges, at a cost point and operational simplicity that makes it deployable across the full spectrum of India's healthcare system, from tertiary hospitals in metropolitan cities to primary health centres in rural Karnataka.

5.3 RECOMMENDATIONS

5.3.1 Recommendations

1. Integrate AI-based breast cancer screening tools as a complementary first-line screening option alongside traditional mammography. Given the demonstrated comfort advantage (mean score 4.39/5) and positive patient perception (87.1% high comfort), non-invasive AI screening tools can be positioned as the preferred entry point for first-time screeners and younger patients, with traditional imaging reserved for follow-up confirmation of high-risk classifications.

2. Establish clear clinical governance protocols for AI diagnostic tools, including mandatory physician review of all AI-generated risk reports before clinical action. The finding that 79.6% of respondents would trust AI diagnosis without physician confirmation underscores the urgency of building explicit physician-in-the-loop workflows into AI deployment frameworks. Trust in AI should be calibrated and guided, not unconstrained.

3. Invest in clinician education and change management programmes to address physician scepticism and resistance to AI-assisted diagnostics. Building physician champions within hospital departments—clinicians who can demonstrate and vouch for AI tool performance—is the most effective pathway to institutional adoption. Academic medical centres and teaching hospitals should lead this effort through formal training curricula.

4. Engage corporate health programme administrators to include AI-based breast cancer screening in annual employee health check packages. The affordability (66.7% perceive AI as cost-effective) and speed (72.2% agree AI reduces diagnosis time) of AI screening make it well-suited to the corporate wellness context, where minimising employee time and cost is a priority.

5.3.2 Recommendations for NIRAMAI Health Analytix and Health-Tech Companies

1. Accelerate rural deployment through mobile screening units and partnerships with government health programmes (National Cancer Screening Programme, AB-PMJAY). The study found that only 1.9% of respondents were from rural areas, reflecting the current access gap. Deploying Thermalytix mobile units to rural primary health centres—staffed by trained Accredited Social Health Activists (ASHAs) or auxiliary nurses—can directly address the 72.2% of respondents who believe AI can reach low-income populations.

2. Develop a structured patient education and reassurance programme that clearly communicates the supportive role of AI in diagnosis. Given the finding that 79.6% would trust AI without a doctor, marketing communications must explicitly reinforce the physician-AI collaboration model. Patient brochures, digital content, and clinical staff briefings should emphasise that AI risk reports are decision-support tools reviewed and confirmed by a qualified clinician.

3. Pursue insurance reimbursement advocacy with the Insurance Regulatory and Development Authority of India (IRDAI) and major insurers to establish reimbursement codes for AI-based thermal breast screening. The current absence of insurance coverage is a significant barrier to mass adoption, and the 66.7% cost-effectiveness perception among respondents provides a strong consumer demand signal to support this advocacy.

4. Invest in next-generation model development using larger, more diverse, and India-specific clinical datasets, incorporating multimodal data (thermal + clinical history + ultrasound) and enhanced explainability tools. The single substantive improvement suggestion from Q20 — recommending diverse data, multimodal integration, enhanced interpretability, and cross-population validation — is fully aligned with published literature and should be incorporated into the R&D roadmap.

5. Develop and publish India-specific clinical validation data from multi-centre studies across diverse Indian demographics (age, skin tone, BMI distribution, comorbidities). Closing the India-specific evidence gap is critical for gaining clinician trust, securing regulatory approvals in new markets, and demonstrating fitness-for-purpose to Indian government health programmes.

5.3.3 Recommendations for Government and Policy Makers

1. Expedite the development and publication of clear SaMD (Software as a Medical Device) regulatory guidelines by CDSCO, providing a defined approval pathway for AI diagnostic tools with risk-proportionate evidence requirements. Regulatory certainty will reduce commercialisation risk for health-tech companies and accelerate the availability of approved AI tools in the Indian healthcare market.

2. Include AI-based non-invasive breast cancer screening in the National Cancer Screening Programme and in the benefit packages of Ayushman Bharat Pradhan Mantri Jan Arogya Yojana (AB-PMJAY), so that low-income women have access to affordable AI screening without out-of-pocket costs. The study's finding that 72.2% believe AI can improve access for low-income groups, combined with NIRAMAI's documented per-screening cost advantage, makes a strong economic case for this inclusion.

3. Fund the creation of India-specific breast cancer diagnostic datasets through ICMR and the National Cancer Registry Programme, to enable AI companies to train and validate models on representative Indian clinical data. Government-facilitated data infrastructure is a public good that will yield significant returns in improved diagnostic quality and reduced cancer mortality.

5.3.4 Recommendations for Future Research

1. Conduct a longitudinal study tracking changes in AI screening awareness, adoption, and trust over a three-to-five year period, to measure the impact

of expanded deployment, regulatory approvals, and insurance inclusion on screening behaviour.

2. Extend the primary survey to a larger, more geographically representative sample—including rural and semi-urban populations—to obtain findings that are generalisable across India's diverse healthcare landscape rather than limited to the educated urban youth demographic of the current study.

3. Undertake a health economics analysis quantifying the system-level cost savings achievable through AI-assisted early breast cancer detection at scale, providing evidence for insurance reimbursement policy decisions and government procurement justifications.

5.4 Management Lessons Learned

5.4.1 Project Planning and Structured Problem-Solving

This internship project reinforced the fundamental importance of structured problem-solving in management practice. The process of defining the research problem precisely, reviewing existing literature to identify gaps, formulating specific and measurable objectives, designing a rigorous data collection instrument, and systematically analysing results mirrors the analytical frameworks used in management consulting, business strategy, and data-driven decision-making. The discipline of following this structured process—resisting the temptation to jump to conclusions before the evidence is collected and analysed—is a habit of mind that is directly transferable to management roles in any industry.

Time management was a critical lesson. Executing a multi-stage research project—from problem formulation through data collection, analysis, and report writing—within a twelve-week internship period required careful planning, milestone-setting, and adherence to a project timeline. Delays in the survey distribution phase required a rapid adjustment of the data collection strategy, demonstrating the importance of building contingency buffers into project plans and being agile in response to execution challenges.

5.4.2 Understanding the Intersection of Technology and Business

One of the most valuable lessons from this project is a deepened understanding of how advanced technology capabilities—in this case, machine learning and AI—translate (or fail to translate) into market adoption and social impact. The study's finding that 83.3% awareness of AI diagnostics has not yet translated into corresponding adoption rates illustrates a fundamental principle of technology management: technological capability is necessary but not sufficient for commercial success. Market adoption requires the simultaneous alignment of product-market fit, affordability, distribution infrastructure, regulatory approval, reimbursement coverage, and clinician acceptance.

This insight has direct management implications. As a future management professional working in or adjacent to the technology sector, the lesson is clear: strategy must address the full ecosystem of adoption barriers, not only the technical performance of a product. The most accurate ML model in the world will not save lives if it is not deployed in the right settings, at the right price, with the right clinical governance framework.

5.4.3 Stakeholder Management and Cross-Functional Collaboration

Observing NIRAMAI's organisational model and the multi-stakeholder nature of healthcare AI deployment provided practical insight into the complexity of stakeholder management. The successful deployment of an AI diagnostic tool requires the alignment of diverse stakeholders with different priorities: clinicians (who prioritise accuracy, interpretability, and clinical workflow integration), patients (who prioritise comfort, privacy, and cost), hospital administrators (who prioritise return on investment and operational efficiency), regulators (who prioritise safety and evidence), and investors (who prioritise growth and commercial viability).

Managing these stakeholders requires not only technical competence but also communication skill, empathy, and the ability to translate complex technical information into language that resonates with non-technical audiences. The skills of framing, persuasion, and stakeholder mapping—core competencies of MBA education—are as important in the health-tech space as they are in any other industry.

5.4.4 Data-Driven Decision Making

This project provided hands-on experience in the full cycle of data-driven decision-making: from research design and instrument development, through data collection and cleaning, to statistical analysis, visual presentation, and insight generation. Working with real survey data—including the messiness of open-ended responses, the challenge of ensuring respondent representativeness, and the discipline of letting data speak rather than imposing pre-conceived conclusions—built a practical appreciation for both the power and the limitations of quantitative research.

The lesson that data quality is as important as analytical sophistication was particularly salient. A perfectly executed statistical analysis of a biased or unrepresentative dataset will produce confident but misleading conclusions. As a future manager, the commitment to ensuring that the data underlying business decisions is accurate, representative, and collected through rigorous processes is a professional responsibility that this project has embedded as a core value.

5.4.5 Ethical Responsibility in Technology and Research

This project heightened awareness of the ethical dimensions of both AI deployment in healthcare and academic research. On the AI deployment side, the finding that 79.6% of respondents would trust an AI diagnosis without physician oversight highlighted the ethical imperative for responsible communication of AI capabilities and limitations. Overstating AI diagnostic accuracy, understating the importance of physician review, or deploying AI tools in settings without adequate clinical governance frameworks could directly harm patients—a consequence far more serious than any commercial damage.

On the research side, the obligation to acknowledge limitations honestly, cite sources accurately, report findings without selective distortion, and avoid plagiarism in academic work are not merely formal rules but expressions of the intellectual integrity that is the foundation of trustworthy research. These ethical commitments, practised and internalised through this project, are skills that will serve well throughout a professional career.

5.5 Closing Statement

This study set out to investigate the awareness, perceptions, and attitudes of respondents toward AI/ML-based breast cancer screening, with specific reference to NIRAMAI Health Analytix's Thermalytix technology. The findings, drawn from a primary survey of 54 respondents, reveal a landscape of strong and growing public support for AI diagnostics—grounded in high awareness, positive direct experience, and widespread recognition of the technology's potential for improving health equity and reducing the economic burden of late-stage cancer diagnosis.

The study has demonstrated that the barriers to AI adoption in breast cancer screening are not technological—the technology is clinically validated, affordable, and operationally deployable—but structural: gaps in insurance reimbursement, regulatory clarity, physician education, and rural infrastructure. Addressing these structural barriers through coordinated action by health-tech innovators, healthcare providers, government health agencies, and academic researchers is the defining challenge and opportunity for AI-driven healthcare in India over the next decade.

The experience of conducting this project—designing a research study, collecting and analysing data, reviewing literature, studying a real-world company, and synthesising findings into actionable recommendations—has provided an invaluable integration of the management and analytical skills developed through the MBA programme. It has reinforced the conviction that data science and management are not separate disciplines but deeply complementary ones, and that the most impactful work lies at their intersection.

Bibliography

Adewole, K. S., Akande, N. O., Adewumi, A. O., Kayode, A. A., & Garba, E. J. (2021). Breast cancer prediction using machine learning approach. Computational Intelligence and Neuroscience, 2021, 1–14. https://doi.org/10.1155/2021/5528622

Asri, H., Mousannif, H., Al Moatassime, H., & Noel, T. (2016). Using machine learning algorithms for breast cancer risk prediction and diagnosis. Procedia Computer Science, 83, 1064–1069.

Barnett, A. J., Bhimani, J., Gao, C., Chen, X., & Rudin, C. (2021). Interpretable mammographic image classification using case-based reasoning and deep learning. NPJ Digital Medicine, 4(1), 1–11.

Bharat, A., Pooja, N., & Reddy, R. A. (2021). Using machine learning algorithms for breast cancer risk prediction and diagnosis. Applied Sciences, 11(9), 3771.

Grand View Research. (2025). Digital health market size, share & trends analysis report. Grand View Research. https://www.grandviewresearch.com/industry-analysis/digital-health-market

Indian Council of Medical Research (ICMR). (2024). National cancer registry programme report 2024. ICMR. https://ncrpindia.org

Kourou, K., Exarchos, T. P., Exarchos, K. P., Karamouzis, M. V., & Fotiadis, D. I. (2015). Machine learning applications in cancer prognosis and prediction. Computational and Structural Biotechnology Journal, 13, 8–17.

Ministry of Health and Family Welfare. (2017). National health policy 2017. Government of India. https://www.mohfw.gov.in

NASSCOM & Deloitte. (2025). AI in healthcare: India's opportunity. NASSCOM. https://nasscom.in

NIRAMAI Health Analytix. (2025). Thermalytix product overview and clinical evidence. NIRAMAI. https://www.niramai.com

NITI Aayog. (2018). National strategy for artificial intelligence. NITI Aayog, Government of India. https://niti.gov.in

Salama, G. I., Abdelhalim, M. B., & Zeid, M. A. E. (2012). Breast cancer diagnosis on three different datasets using multi-classifiers. International Journal of Computer and Information Technology, 1(1), 36–43.

UCI Machine Learning Repository. (2023). Breast cancer Wisconsin (diagnostic) data set. University of California, Irvine. https://archive.ics.uci.edu/ml/datasets/Breast+Cancer+Wisconsin+(Diagnostic)

World Health Organization. (2024). Breast cancer. WHO. https://www.who.int/news-room/fact-sheets/detail/breast-cancer

Annexures

Research questions

1. **Name of the Respondents**
2. **what is your age group ?**
   a) 18–20
   b) 21-25
   c) 26 - 35
   d) Above 35
3. **What is your gender?**
   a) Male
   b) Female
   c) Prefer not to say
   d) Other
4. **What is your current level of education?**
   a) UG
   b) PG
   c) Professional Course
   d) Other:
5. **Relationship Status**
   a) Single
   b) Married
   c) Other:
6. **Location**
   a) City
   b) Town
   c) Rural Areas
7. **Are you aware of AI/ML technologies used in breast cancer detection?**
   a) Yes
   b) No
8. **Have you heard of non-invasive screening methods like thermal imaging (e.g., Niramai)?**
   a) Yes
   b) No

9. **Have you undergone breast cancer screening?**
    a) Yes (Traditional methods like mammogram/ultrasound)
    b) Yes (AI-based method)
    c) No
10. **If yes, which method did you prefer?**
    a) Traditional
    b) AI/ML Methode
    c) Both
11. **Comfort during screening process**
    a) 1
    b) 2
    c) 3
    d) 4
    e) 5
12. **Accuracy and trust in results**
    a) 1
    b) 2
    c) 3
    d) 4
    e) 5
13. **AI/ML-based screening is more comfortable than traditional methods**
    a) Agree
    b) Disagree
    c) Neutral
14. **AI/ML reduces diagnosis time compared to traditional methods**
    a) Agree
    b) Disagree
    c) Neutral

15. **AI/ML is more cost-effective than traditional screening**
    a) Agree
    b) Disagree
    c) Neutral
16. **Would you trust AI-based diagnosis without a doctor's confirmation?**
    a) Yes
    b) No
17. **Would you recommend AI-based breast cancer screening to others?**
    a) No
    b) Yes
18. **Do you believe AI screening can make cancer detection accessible to low-income groups?**
    a) Yes
    b) No
    c) Maybe
19. **Do you think AI/ML reduces overall hospital costs?**
    a) Yes
    b) No
    c) Maybe
20. **What improvements would you suggest for AI-based breast cancer detection?**